

VIETNAM NATIONAL UNIVERSITY - HO CHI MINH CITY

UNIVERSITY OF ECONOMICS AND LAW

FACULTY OF ECONOMIC MATHEMATICS



FINAL PROJECT REPORT

MACHINE LEARNING

**PREDICTING STUDENTS' MENTAL HEALTH
CONDITIONS USING MACHINE LEARNING**

Lecturer: M.Sc. Trương Quang Nhật

Course Code: 252BMA400602

Student Name: Bùi Thị Phước Trâm

Student ID: K234131574

Ho Chi Minh City, March 2026

ACKNOWLEDGEMENT

I would like to extend our deepest gratitude to **M.Sc. Trương Quang Nhật** for his invaluable guidance and dedicated mentorship throughout the duration of this project. His profound expertise, alongside his insightful critiques and constructive suggestions, has been instrumental in refining our methodology and enhancing the overall quality of this report.

I am profoundly grateful to the **Faculty of Economic Mathematics**. The rigorous academic environment and the foundational knowledge provided by the Faculty have served as a solid cornerstone for my research.

While I have poured our utmost dedication and effort into this study, I recognize that certain limitations and oversights may still exist due to my burgeoning experience. I, therefore, sincerely welcome any comments and feedback from my esteemed lecturers and readers. Such insights will undoubtedly serve as a precious asset, helping me to broaden my intellectual horizons and perfect my skills in future endeavors.

Best Regards./.

COMMITMENT

I hereby solemnly declare that this report is the result of my own independent research and intellectual labor. All content, analytical results, and source code presented herein are authentic and have not been previously submitted, in part or in whole, for any other academic credit or coursework. Furthermore, all external sources, datasets, and references have been meticulously cited and credited in accordance with academic standards. I assume full and sole responsibility for the integrity, accuracy, and truthfulness of the information provided in this report.

Ho Chi Minh City, March 2026

Declared by,

Bùi Thị Phước Trâm – K234131574

CONTENTS

ACKNOWLEDGEMENT	ii
COMMITMENT	iii
CONTENTS	iv
LIST OF FIGURES	vii
LIST OF TABLES	viii
LIST OF ABBREVIATIONS.....	ix
ABSTRACT	x
CHAPTER 1: INTRODUCTION.....	1
1.1. Background and Rationale	1
1.2. Problem Statement	1
1.3. Objectives	2
1.4. Research Questions	2
1.5. Structure of the Thesis.....	2
CHAPTER 2: LITERATURE REVIEW	4
2.1. Student Mental Health.....	4
2.1.1. Major Psychological Conditions.....	4
2.1.2. Influencing Factors (Determinants)	4
2.2. Machine Learning in Healthcare	5
2.2.1. Classification Tasks	5
2.2.2. Predictive Analytics	5
2.2.3. Machine Learning Pipeline.....	5
2.3. Common Models	6
2.4. Related Studies	7
2.4.1. Multidimensional Feature Integration	7
2.4.2. Comparative Analysis of Algorithmic Performance.....	8
2.4.3. Impact of Academic Specialization on Mental Health	8

2.4.4. Addressing the "Early Warning" Gap	8
2.5. Research Gap	9
CHAPTER 3: METHODOLOGY	10
3.1. Problem Formulation	10
3.2. Model Selection	10
3.2.1. Logistics Regression (LR)	11
3.2.2. Support Vector Machine (SVM)	11
3.2.3. Random Forest (RF)	12
3.3. Workflow	12
3.4. Dataset Description	13
3.5. Variables Description	14
3.6. Data Overview	15
3.7. Data Preprocessing	15
3.7.1. Cleaning	15
3.7.2. Exploratory Data Analysis (EDA)	15
3.7.2.1. Distribution of Student Depression Risk (Target)	15
3.7.2.2. Histogram of Numerical Variables	16
3.7.2.3. Boxplot of Numerical Variables	18
3.7.2.4. Comparison of Numerical Variables by Target	19
3.7.2.5. Distribution of Demographic and Background Variables	20
3.7.2.6. Distribution of Lifestyle and Behavioral Variables	21
3.7.2.7. Distribution of Health and Support System Variables	23
3.7.2.8. Distribution of Demographic and Background Variables by Target	25
3.7.2.9. Distribution of Lifestyle and Behavioral Variables by Target	26
3.7.2.10. Distribution of Health and Support System Variables by Target	28

3.7.2.11. Correlation of Numerical Variables with Target	29
3.7.3. Transformation.....	30
3.7.4. Splitting.....	31
3.7.5. Feature Selection	32
3.7.5.1. Chi-Square Test.....	32
3.7.5.2. Random Forest Feature Importance.....	34
CHAPTER 4: RESULTS AND DISCUSSION	38
4.1. Overview of Experimental Setup	38
4.2. Results	39
4.2.1. Performance on the Selected Dataset	40
4.2.2. Performance on the Full Dataset.....	41
4.2.3. Comparison of Feature Sets and Models	42
4.2.4. Feature Importance Analysis using Logistic Regression.....	43
4.2.5. Classification Performance and Error Analysis.....	45
4.2.6. ROC Curve	47
4.2.7. The Robustness of the Logistic Regression.....	50
4.3. Discussion	50
CHAPTER 5: CONCLUSION.....	52
5.1. Summary.....	52
5.2. Recommendations	52
5.2.1. For the University and Faculty Administration	52
5.2.2. For the Student Counseling Center.....	53
5.3. Limitations and Future Research.....	53
REFERENCES	54

LIST OF FIGURES

Figure 3.1. Research Workflow	13
Figure 3.2. Distribution of Student Depression Risk	16
Figure 3.3. Distribution of Numerical Variables	17
Figure 3.4. Boxplot of Numerical Variables.....	18
Figure 3.5. Boxplot of Numerical Variables by Target.....	19
Figure 3.6. Countplot of Demographic and Background Variables	21
Figure 3.7. Countplot of Lifestyle and Behavioral Variables.....	22
Figure 3.8. Countplot of Health and Support System variables.....	24
Figure 3.9. Demographic and Background Variables by Depression Status.....	25
Figure 3.10. Lifestyle and Behavioral Variables by Depression Status	27
Figure 3.11. Health and Support System Variables by Depression Status	28
Figure 3.12. Correlation with Depression Status	29
Figure 3.13. Feature Importance from Random Forest Classifier.....	36
Figure 4.1. Confusion Matrix – Logistic Regression	46
Figure 4.2. Operating Characteristic (ROC) Curve.....	49

LIST OF TABLES

Table 3.1. Variables Description.....	14
Table 3.2. Descriptive Statistics	15
Table 3.3. Chi-Square Results (Train Set Only).....	33
Table 3.4. Feature Importance from Random Forest Classifier	35
Table 4.1. Performance on the Selected dataset	40
Table 4.2. Performance on the Full Dataset	41
Table 4.3. Detailed Feature Analysis (Logistic Regression)	44

LIST OF ABBREVIATIONS

AUC	Area Under the Curve
CDSS	Clinical Decision Support Systems
CGPA	Cumulative Grade Point Average
EDA	Exploratory Data Analysis
FN	False Negative
FP	False Positive
IQR	Interquartile Ranges
LR	Logistic Regression
ML	Machine Learning
RBF	Radial Basis Function
RF	Random Forest
ROC	Receiver Operating Characteristic
STEM	Science - Technology - Engineering - Mathematics
SVM	Support Vector Machine
TN	True Negative
TP	True Positive

ABSTRACT

This study aims to analyze and predict students' mental health status using machine learning techniques. The dataset, collected from the Students' Mental Health Survey available on Kaggle, includes demographic, academic, and psychological factors influencing mental well-being. After data preprocessing and feature engineering, three classification models - Logistic Regression, Support Vector Machine (SVM), and Random Forest - are applied to classify students into low-risk and high-risk groups for depression. The models are evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score. The results provide a comparative analysis of model performance and identify the most effective approach for predicting students' mental health status. This study contributes to early detection efforts and supports data-driven decision-making in student mental health management.

CHAPTER 1: INTRODUCTION

1.1. Background and Rationale

In the contemporary global landscape, mental health has emerged as a cornerstone of public health, with the World Health Organization, 2022 reporting that nearly one billion individuals live with mental disorders, positioning psychological well-being as a prerequisite for sustainable socioeconomic development. Within this broad spectrum, the university student population represents a uniquely vulnerable demographic, navigating a high-stakes transition marked by intense academic competition, evolving social identities, and increasing financial instability. This period often functions as a "pressure cooker" where the relentless drive for academic excellence and professional success intersects with chronic sleep deprivation and the loss of traditional family support systems (American Psychological Association, 2022). The consequences of untreated psychological distress - ranging from persistent depression and anxiety to severe panic disorders - extend far beyond individual suffering, directly contributing to academic attrition, impaired cognitive function, and long-term socioeconomic disadvantages (Eisenberg et al., 2013). Despite the rising prevalence of these conditions, a critical "Identification Gap" remains a pervasive challenge in higher education; social stigma and institutional resource constraints often leave psychological red flags undetected until they escalate into a crisis (Lipson et al., 2022). Most existing support frameworks remain inherently reactive, intervening only after significant functional impairment has occurred, rather than providing the proactive, preventative care necessary for early intervention. There is an urgent and undeniable need for scalable, efficient, and non-invasive methods to identify at-risk students at an early stage, ensuring the health, resilience, and productivity of the next generation of professionals.

1.2. Problem Statement

While institutional awareness of student mental health has increased significantly, the accurate and timely identification of at-risk individuals continues to be a formidable challenge. Current screening protocols rely largely on periodic self-reported surveys and manual clinical evaluations, which are inherently prone to subjective and social desirability

bias, as well as inter-rater inconsistency (Shatte et al., 2019). These traditional methods are labor-intensive and often fail to scale effectively across large student populations, leading to a significant "detection lag" where students only receive support after symptoms have reached a critical threshold. From a technical perspective, conventional statistical frameworks often struggle to account for the complex, non-linear interactions between a diverse array of academic, lifestyle, and socio-economic variables that contribute to psychological distress (James et al., 2021). There is a critical gap in providing a proactive, objective, and scalable assessment framework. This study addresses this need by developing a data-driven approach that leverages machine learning techniques to enhance the early identification of students at risk of mental health complications, thereby facilitating more timely and effective institutional interventions.

1.3. Objectives

The objectives of this study are fourfold. First, it aims to analyze the student mental health dataset in order to understand the overall patterns and characteristics of the data. Second, the study seeks to identify the key factors that influence mental health conditions among students, such as stress, anxiety, and depression levels. Third, it focuses on building predictive models using various machine learning algorithms to classify or predict mental health outcomes. Finally, the study compares the performance of these models to determine which algorithm provides the most accurate and reliable results.

1.4. Research Questions

This study aims to answer two main research questions: what factors significantly influence students' mental health, and which machine learning model provides the best prediction performance in analyzing and predicting students' mental health conditions.

1.5. Structure of the Thesis

The research consists of 65 pages, 15 figures, and 7 tables. In addition to the introduction, conclusion, list of abbreviations, list of tables and figures, references, and appendices, the main content of the report is structured into the following chapters:

Chapter 1: Introduction

Chapter 2: Literature Review

Chapter 3: Methodology

Chapter 4: Results and Discussion

Chapter 5: Conclusion

CHAPTER 2: LITERATURE REVIEW

2.1. Student Mental Health

In recent years, the mental health of university students has emerged as a significant global public health concern. Academic environments, characterized by high-stakes evaluations and social transitions, often exacerbate psychological vulnerabilities (American Psychological Association, 2022). Common mental health conditions observed in this demographic include depression, anxiety, and panic disorders, which frequently exhibit comorbidity - the simultaneous presence of two or more conditions.

2.1.1. Major Psychological Conditions

According to the (World Health Organization, 2023) depression is a leading cause of disability worldwide. In a student context, it manifests as persistent sadness, anhedonia (loss of interest), and impaired cognitive function, which directly correlates with decreased academic self-efficacy and higher dropout rates.

Anxiety involves chronic, excessive worry and physical tension, often triggered by academic performance pressure (Lipson et al., 2022). In severe cases, this may escalate into panic attacks - sudden, intense episodes of fear accompanied by autonomic symptoms such as tachycardia (rapid heartbeat) and dyspnea (shortness of breath).

2.1.2. Influencing Factors (Determinants)

The etiology of student mental health issues is multifactorial. Recent longitudinal studies indicate that environmental and lifestyle factors are primary predictors of psychological outcomes:

Poor sleep quality and inadequate physical activity are strongly linked to increased levels of cortisol and emotional instability.

Financial instability and academic workload remain constant stressors that can trigger or worsen pre-existing conditions.

Robust social support systems act as a critical buffer, mitigating the impact of stress and fostering resilience among students (Eisenberg et al., 2013).

2.2. Machine Learning in Healthcare

Machine Learning (ML) has emerged as a transformative force in modern healthcare, providing sophisticated Clinical Decision Support Systems (CDSS) to analyze complex medical and psychological datasets (Topol, 2019). By leveraging computational power, ML supports data-driven decision-making, allowing for a more nuanced understanding of human health that traditional statistical methods might overlook.

In the field of mental health, ML techniques are increasingly utilized to identify hidden patterns within behavioral and social data. According to Shatte et al. (2019), ML excels in risk stratification, helping practitioners distinguish between various levels of psychological distress.

2.2.1. Classification Tasks

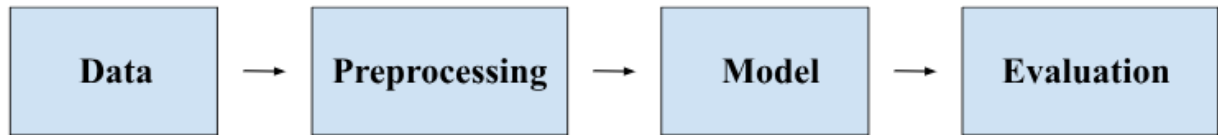
Among various applications, classification is the most prevalent task in supervised learning. It involves assigning observations to predefined categories based on input features. In mental health research, classification models are instrumental for the early detection of disorders. By categorizing individuals into low, medium, or high-risk groups for conditions such as stress, anxiety, or depression, ML enables timely and personalized interventions, which are crucial for preventing long-term psychological deterioration.

2.2.2. Predictive Analytics

Predictive analytics represents another critical pillar, where ML models utilize historical and real-time data to forecast future health outcomes (Rajkomar et al., 2019). In the context of student mental health, these models can estimate the likelihood of psychological issues by analyzing a multifactorial range of variables, including academic performance, lifestyle choices, and social interaction patterns. This proactive approach allows educational institutions to provide support before a crisis occurs.

2.2.3. Machine Learning Pipeline

The machine learning process typically follows a structured process:



Machine Learning pipeline

A critical consideration in building ML models is managing overfitting and underfitting. Overfitting occurs when a model learns the training data too closely, including noise, resulting in poor performance on new data. Underfitting happens when a model is too simple to capture underlying patterns in the data. Balancing these issues is essential for creating models that generalize well to unseen data.

2.3. Common Models

For the binary classification task in this study, three distinct machine learning algorithms are implemented: Logistic Regression, Support Vector Machine (SVM), and Random Forest. These models are selected to provide a comprehensive evaluation by comparing a traditional statistical baseline with advanced classification and ensemble methods, ensuring a balance between interpretability and predictive accuracy.

Logistic Regression is employed as the baseline model due to its efficiency in binary outcomes. Beyond its simplicity, LR is particularly valuable for this research because it allows for the calculation of Odds Ratios from the regression coefficients. This enables a precise quantification of how specific groups of features - such as academic background or socio-economic status - directly increase or decrease the likelihood of a student being classified as "High Risk". The model's probabilistic output provides a flexible foundation for educational administrators to adjust intervention thresholds based on institutional resources and the severity of predicted risks.

SVM is selected for its robust performance in identifying the optimal separating hyperplane within the 15-dimensional feature space. The core objective of SVM is to maximize the geometric margin between the two classes, ensuring that the decision boundary is as far as possible from the nearest data points, known as Support Vectors. This model is exceptionally effective for mental health datasets where psychological scores and

lifestyle metrics often exhibit overlapping distributions. By focusing on these boundary cases, SVM provides a highly stable classification framework that remains effective even when the transition between "Low Risk" and "High Risk" states is subtle and non-linear.

Random Forest is included as an ensemble learning representative designed to achieve superior predictive performance by combining the outputs of 200 individual decision trees. The model utilizes Bootstrap Aggregating (Bagging) and Majority Voting to arrive at the final classification, which significantly reduces the variance typically associated with single-model approaches. The suitability of RF for this study lies in its inherent ability to capture complex, non-linear interactions across the 15 features without requiring explicit feature engineering. It effectively evaluates the compounded impact of multiple stressors - such as the synergy between high academic load and low social support - providing a more nuanced assessment than linear methods. The internal ranking of feature importance offers a secondary layer of validation for the primary risk factors identified across the various categories of student data.

2.4. Related Studies

The application of Machine Learning (ML) in psychological health has seen a surge in recent years, with researchers shifting from descriptive statistics to predictive modeling to address the growing mental health crisis in higher education. This section reviews key empirical studies that have shaped the current understanding of ML-driven mental health assessment.

2.4.1. Multidimensional Feature Integration

A pivotal study published in (Li et al., 2025) examined the efficacy of ML in identifying mental health risks among university students by integrating behavioral and psychological markers. The researchers found that models achieved significantly higher accuracy when combining academic workload data with lifestyle variables such as sleep hygiene and physical activity. Their findings emphasize that stress and anxiety scores are

the most potent predictors of long-term psychological outcomes, a conclusion that aligns with the feature selection strategy adopted in this research.

2.4.2. Comparative Analysis of Algorithmic Performance

The debate between model complexity and interpretability is a recurring theme in recent literature. A comprehensive review in Yin et al., (2024) compared traditional statistical models with ensemble architectures. The study revealed that while Random Forest and Support Vector Machines (SVM) excel in capturing non-linear relationships in complex datasets, Logistic Regression remains a highly competitive benchmark due to its stability and the transparency of its coefficients. This research justifies the use of a multi-model approach to ensure both high predictive performance and clinical explainability.

2.4.3. Impact of Academic Specialization on Mental Health

A large-scale survey was analyzed in the Lipson et al., (2022) focused on how specific academic disciplines influence student well-being. The study identified that students in high-intensity STEM fields, particularly Computer Science and Engineering, reported higher levels of burnout and anxiety compared to their peers in social sciences. This specialized focus supports the inclusion of "Course Type" as a categorical feature in predictive models to account for discipline-specific stressors.

2.4.4. Addressing the "Early Warning" Gap

Beyond technical metrics, the Li et al., (2025) emphasized the systemic necessity of data-driven approaches. Their report identified a critical "Early Warning" gap in educational institutions, where subtle psychological shifts often go unnoticed by traditional advisory systems. They advocate for the deployment of automated ML screening tools to provide proactive interventions, reinforcing the practical significance and urgency of the predictive framework developed in this thesis.

2.5. Research Gap

Although previous studies have demonstrated the effectiveness of machine learning in mental health prediction, several limitations remain. Many studies focus on general populations rather than specifically targeting student datasets. Some research does not incorporate a comprehensive set of lifestyle and behavioral variables. Limited studies provide a systematic comparison between multiple machine learning models on the same dataset.

This study aims to address these gaps by focusing on a student-specific dataset, incorporating multiple relevant features, and comparing the performance of different machine learning models.

CHAPTER 3: METHODOLOGY

3.1. Problem Formulation

In this research, the problem is formulated as a binary classification task with the objective of predicting students' mental health status. Specifically, the model aims to classify whether a student is at risk of mental health issues or not, based on observed characteristics.

The target variable is constructed from mental health indicators, particularly the Depression Status. This target variable classifies students into low-risk and high-risk groups for depression; specifically, a Depression Score of 3 or higher indicates high risk, while a score below 3 indicates low risk. A threshold is applied to transform this variable into a binary outcome, where one class represents students with potential mental health risk and the other represents students without significant risk. This transformation allows the problem to be addressed using classification algorithms.

The input variables consist of a set of features describing each student, including demographic factors (e.g., Age, Gender), academic performance (e.g., CGPA, Semester_Credit_Load), and lifestyle-related variables (e.g., Sleep_Quality, Physical_Activity, Diet_Quality, and Social_Support). These features are used as predictors to capture patterns associated with students' mental health conditions.

Overall, the goal of the model is to learn the relationship between input features and the target variable, thereby enabling accurate classification of students into risk and non-risk groups. This formulation provides a foundation for applying and comparing different machine learning algorithms in subsequent sections.

3.2. Model Selection

In this research, several supervised machine learning algorithms are implemented to classify and predict Depression Status. According to James et al. (2021), classification is a fundamental task in statistical learning where the goal is to predict a qualitative response for an observation. The following models - Logistic Regression, Support Vector Machine

(SVM), and Random Forest - were selected due to their proven effectiveness in handling structured healthcare and social science data.

3.2.1. Logistics Regression (LR)

Logistic Regression is a foundational statistical model used for binary classification. It estimates the probability that a given observation belongs to a specific category. Unlike linear regression, which predicts continuous values, LR uses the Sigmoid (logistic) function to ensure the output remains within the range of $[0, 1]$ (Cox, 1958). The probability is calculated as:

$$P(y = 1|x) = \sigma(z) = \frac{1}{1 + e^{-z}}$$

Where z is the linear combination of input features:

$$z = \beta_0 + \beta_1x_1 + \beta_2x_2 + \dots + \beta_nx_n$$

The coefficients (β) represent the weight of each feature, allowing for high interpretability through Odds Ratios ($OR = e^{\beta}$). This model is particularly valuable in healthcare for identifying significant risk factors (Hastie et al., 2009).

3.2.2. Support Vector Machine (SVM)

SVM is a robust supervised learning algorithm designed to find the optimal hyperplane that separates data points of different classes with the maximum margin (Cortes & Vapnik, 1995). The data points located on the edge of this margin are called Support Vectors, which are critical in defining the decision boundary.

The decision function for a linear SVM is:

$$f(x) = \text{sign}(w^T x + b)$$

Where w is the weight vector and b is the bias. To handle the complex, non-linear psychological patterns inherent in student survey data, SVM employs the Kernel Trick.

This mathematical transformation projects the data into a higher-dimensional space where a linear separation becomes possible:

$$K(x_i, x_j) = \phi(x_i)^T \phi(x_j)$$

By using the Radial Basis Function (RBF) kernel, the model can identify intricate clusters of high-risk individuals that are not linearly separable in the original feature space.

3.2.3. Random Forest (RF)

Random Forest is an ensemble learning method that constructs a multitude of decision trees during the training phase. It addresses the limitation of individual decision trees - specifically their tendency to overfit - by utilizing Bootstrap Aggregating (Bagging) and feature randomness (Breiman, 2001).

Each tree in the forest is trained on a random subset of the data, and the final prediction is determined by a majority voting mechanism:

$$\hat{Y} = \text{mode}\{T_1(x), T_2(x), \dots, T_B(x)\}$$

Where B is the number of trees and $T_i(x)$ is the output of an individual tree. By combining the results of diverse trees, RF provides higher accuracy and robustness against noise. It provides a measure of Feature Importance, which helps this research identify which academic or lifestyle variables are the most significant predictors of mental health issues. All implementations in this study were conducted using the Scikit-learn library (Pedregosa et al., 2011).

3.3. Workflow

The overall workflow of this study is illustrated in Figure 3.1, showing the process from raw data to model evaluation based on the student mental health dataset.



Figure 3.1. Research Workflow

First, the raw dataset containing students' demographic, academic, and mental health information is collected from Kaggle.

Next, data preprocessing is conducted, including handling missing values and performing exploratory data analysis (EDA) to better understand the data distribution and identify potential issues.

After that, feature engineering is applied, which involves encoding categorical variables into a numerical format and selecting relevant features for modeling. Subsequently, the dataset is split into training and testing sets, and machine learning models, including Logistic Regression, SVM, and Random Forest, are trained on the processed data.

Finally, the performance of the models is evaluated using classification metrics such as accuracy, precision, recall, and F1-score to determine the most effective model for predicting students' mental health status.

3.4. Dataset Description

The dataset used in this project focuses on students' mental health and is designed to support the analysis and prediction of psychological conditions such as stress, anxiety, and depression. The main objective of this dataset is to identify factors influencing students' mental well-being and to provide a foundation for building machine learning models.

The dataset was obtained from Kaggle, a widely used online platform for data science and machine learning. It contains survey-based data collected from students, including information related to their demographic characteristics, academic performance, and mental health conditions.

The subjects of the dataset are students from different academic backgrounds. Each observation represents an individual student and includes variables such as age, gender,

course of study, CGPA, stress level, anxiety score, and depression score. These features are used as inputs for both EDA and predictive modeling.

3.5. Variables Description

The dataset includes various variables related to students' demographics, academic performance, and mental health. A brief description of these variables is presented.

Table 3.1. Variables Description

Variable	Data Type	Description
Age	int64	Age of the student
Course	object	Field of study or academic program
Gender	object	Gender of the student
CGPA	float64	Cumulative Grade Point Average
Stress_Level	int64	Level of stress experienced by the student
Anxiety_Score	int64	Score indicating level of anxiety
Sleep_Quality	object	Self-reported sleep quality
Physical_Activity	object	Level of physical activity
Diet_Quality	object	Quality of diet
Social_Support	object	Level of social support received
Relationship_Status	object	Current relationship status
Substance_Use	object	Use of substances such as alcohol or drugs
Counseling_Service_Use	object	Whether the student uses counseling services
Family_History	object	Family history of mental health issues
Chronic_Illness	object	Presence of chronic illness
Financial_Stress	int64	Level of financial stress
Extracurricular_Involvement	object	Participation in extracurricular activities
Semester_Credit_Load	int64	Number of credits taken per semester
Residence_Type	object	Type of living arrangement
Depression_Status	int64	Indicates presence of depression

3.6. Data Overview

The dataset consists of 7022 observations and 20 variables, providing sufficient data for analysis and modeling.

Descriptive statistics of the numerical variables are presented in Table 3.2. It should be noted that the range of each variable is also included in the table, as reflected by the minimum and maximum values.

Table 3.2. Descriptive Statistics

Variable	Count	Mean	Std.dev	Min	25%	50%	75%	Max
Age	7022	23.003	3.854	18.000	20.000	22.000	25.000	35.000
CGPA	7010	3.491	0.287	2.440	3.290	3.500	3.700	4.000
Stress_Level	7022	2.428	1.638	0.000	1.000	2.000	4.000	5.000
Anxiety_Score	7022	2.300	1.624	0.000	1.000	2.000	4.000	5.000
Financial_Stress	7022	2.453	1.709	0.000	1.000	2.000	4.000	5.000
Semester_Credit_Load	7022	22.011	4.358	15.000	18.000	22.000	26.000	29.000
Depression_Status	7022	0.444	0.497	0.000	0.000	0.000	1.000	1.000

**All numerical values are rounded to three decimal places.*

3.7. Data Preprocessing

3.7.1. Cleaning

Data cleaning was conducted to ensure the dataset's quality and consistency. An initial inspection revealed that missing values were present only in CGPA and Substance_Use, with a relatively small number of observations. Given the limited extent of missing data, imputation was used rather than removal. Specifically, missing values in CGPA were replaced using the median to reduce the influence of extreme values, while missing values in Substance_Use were filled using the mode, as it is a categorical variable.

3.7.2. Exploratory Data Analysis (EDA)

To provide an overview of the dataset, here is a series of charts visualized from the data after the cleaning step.

3.7.2.1. Distribution of Student Depression Risk (Target)

DISTRIBUTION OF STUDENT DEPRESSION RISK

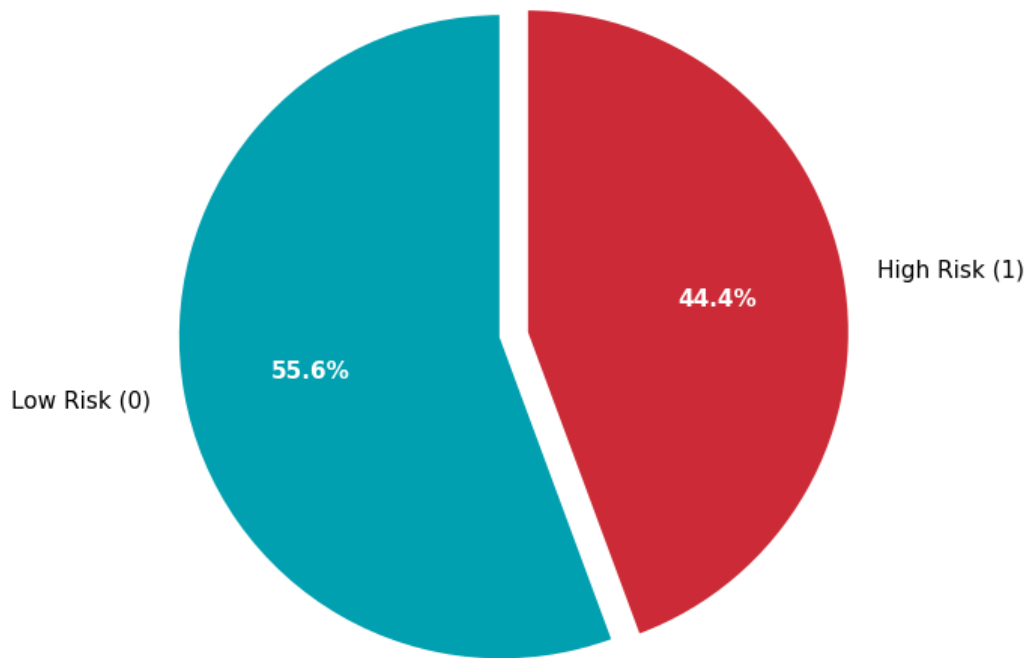


Figure 3.2. Distribution of Student Depression Risk

The pie chart provides a detailed visualization of the target variable, Student Depression Risk, which is categorized into two levels: Low Risk (0) and High Risk (1). Within the student dataset, those at Low Risk constitute the majority at 55.6%, while a substantial 44.4% of the student body is identified as High Risk.

This distribution is remarkably well-balanced, which is ideal for training robust machine learning models as it prevents the algorithm from becoming biased toward either group. But, from an educational and psychological standpoint, the fact that nearly half of the students are at high risk is a significant finding. It underscores the prevalence of mental health challenges among the student population, highlighting an urgent need for academic institutions to implement proactive screening and more accessible mental health support services.

3.7.2.2. Histogram of Numerical Variables

DISTRIBUTION OF NUMERICAL VARIABLES

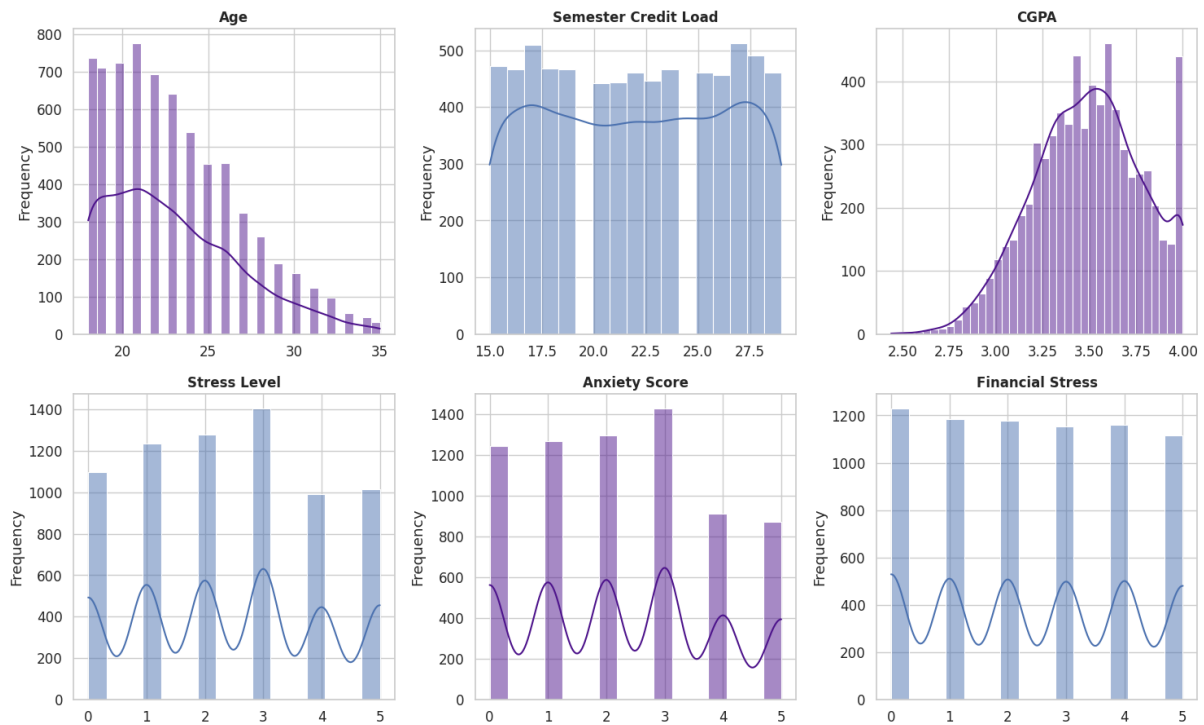


Figure 3.3. Distribution of Numerical Variables

The histograms provide a comprehensive overview of the numerical factors within the dataset.

The Age distribution is notably right-skewed, indicating that the majority of students are in their late teens to early twenties (primarily 18–22 years old), which is typical for an undergraduate population. In terms of academic performance, the CGPA is left-skewed, with a high concentration of students achieving a grade point average between 3.25 and 3.75, suggesting a generally high-performing group. The Semester Credit Load appears relatively uniform across the range of 15 to 29 credits, showing a consistent spread of academic workload among students.

With psychological indicators, Stress Level and Anxiety Score follow similar multi-modal patterns, with frequencies peaking at moderate levels (levels 2 and 3). This suggests that a significant portion of the student body experiences a non-negligible amount of psychological pressure. Meanwhile, Financial Stress is distributed almost evenly across all

levels from 0 to 5, indicating that financial concerns are a persistent issue that affects students across the entire spectrum. Together, these distributions highlight the diverse demographic and mental health profiles that likely contribute to the overall depression risk within the student population.

3.7.2.3. Boxplot of Numerical Variables

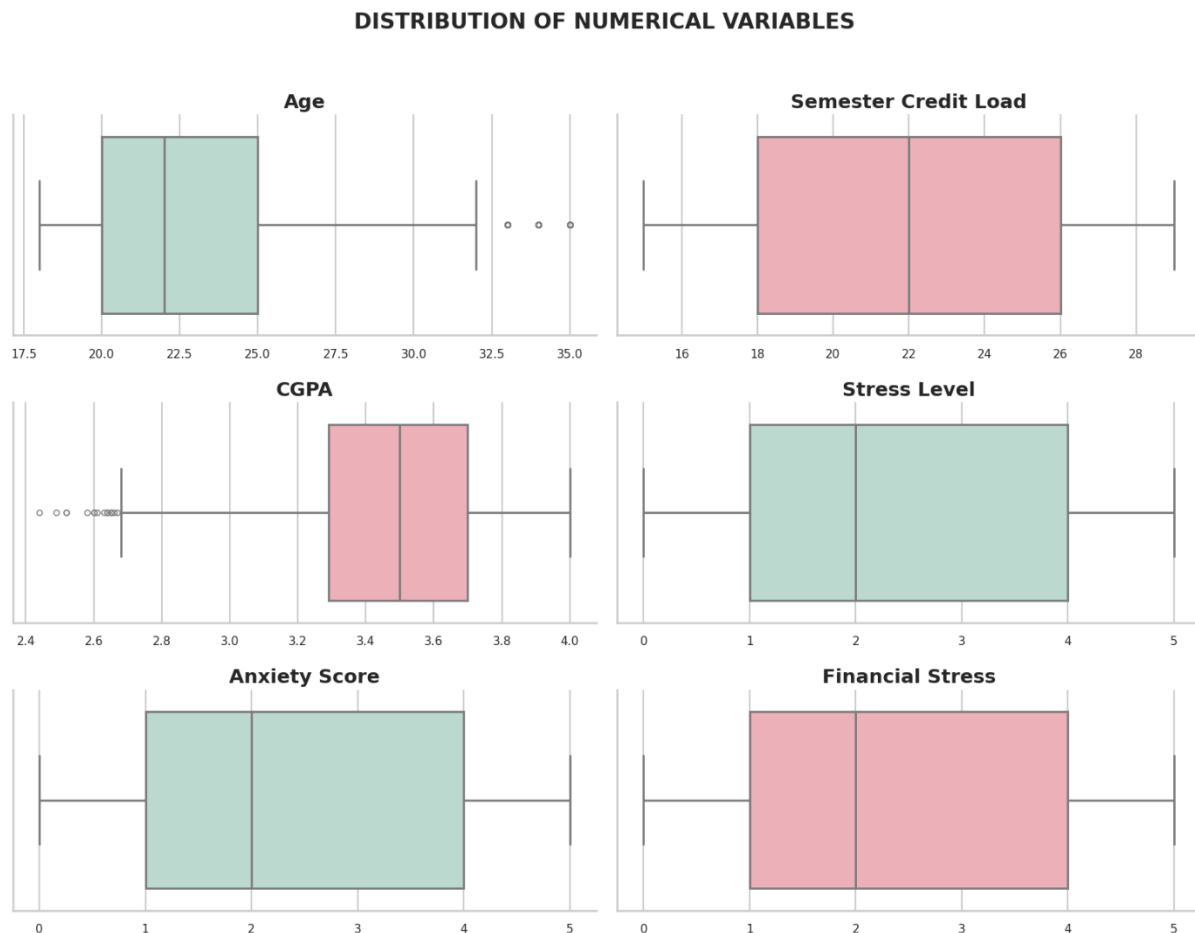


Figure 3.4. Boxplot of Numerical Variables

The Age distribution shows a median of 22, with the majority of students clustered between 20 and 25. The outliers located between 32 and 35 are perfectly logical in a university context. They represent "non-traditional" or mature students who may be returning to education later in life or pursuing a second degree. While they are statistically rare, their presence is important as older students often face different life pressures

compared to their younger peers. These are acceptable age outliers, so there's no need to address them.

CGPA exhibits a left-skewed distribution with a median near 3.5. It contains several low-end outliers (ranging from 2.4 to 2.7). These outliers signify a minority of students who are significantly underperforming compared to the general population. In a study on depression, these outliers are critical "red flags"; a low CGPA can be both a cause and a symptom of mental health struggles, making these data points essential for predictive modeling. Similarly, outliers of the CGPA are also acceptable, as long as they fall within the range of 0 to 4.

The Semester Credit Load is uniformly spread between 15 and 29 credits with no outliers. This is logically consistent with university regulations, which typically impose a minimum and maximum credit limit per semester. Since students cannot exceed these institutional boundaries, the data remains tightly contained within the standard range.

These three variables (Stress Level, Anxiety Score, Financial Stress) show a broad distribution (IQR from 1 to 4) without any outliers. Since these are subjective ratings on a fixed 0–5 scale, all possible responses are considered "normal" within the context of human emotion. The lack of outliers suggests that extreme stress (level 5) or no stress (level 0) is common enough among students that they do not stand out as statistical anomalies.

3.7.2.4. Comparison of Numerical Variables by Target

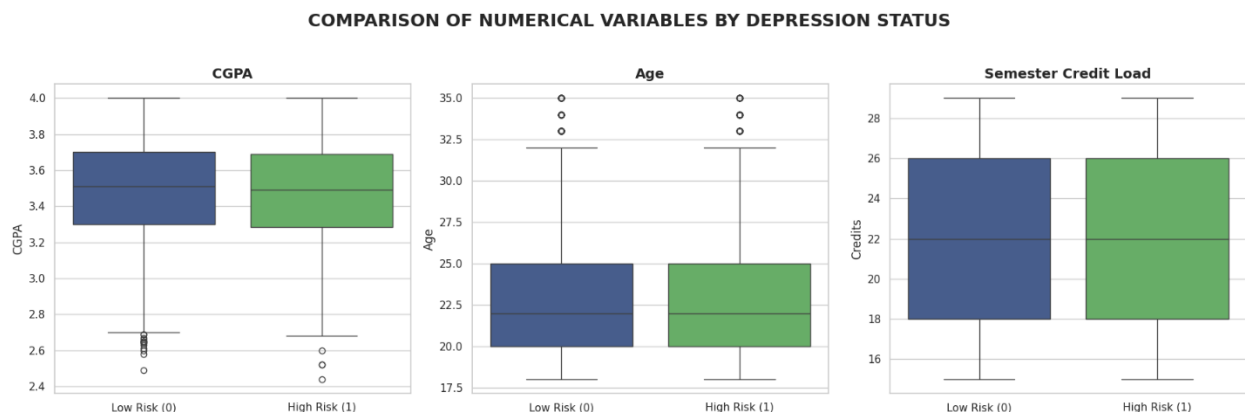


Figure 3.5. Boxplot of Numerical Variables by Target

The side-by-side boxplots reveal a striking uniformity across all three numerical variables when segmented by depression risk. The nearly identical medians and interquartile ranges (IQR) for Age, CGPA, and Semester Credit Load suggest that “High Risk” status is not strongly dictated by these traditional demographic or academic metrics alone.

The overlap in CGPA indicates that high-performing students are just as vulnerable to depression as those with lower grades, debunking the myth that academic success acts as a protective shield. The identical distributions in Age and Semester Credit Load imply that the risk is pervasive across the entire student body, regardless of maturity or workload volume. From a modeling perspective, this “visual similarity” is a crucial finding: it suggests that the primary drivers of depression in this dataset likely reside in psychological or environmental factors rather than simple academic or age-related data.

3.7.2.5. Distribution of Demographic and Background Variables

DISTRIBUTION OF DEMOGRAPHIC AND BACKGROUND VARIABLES

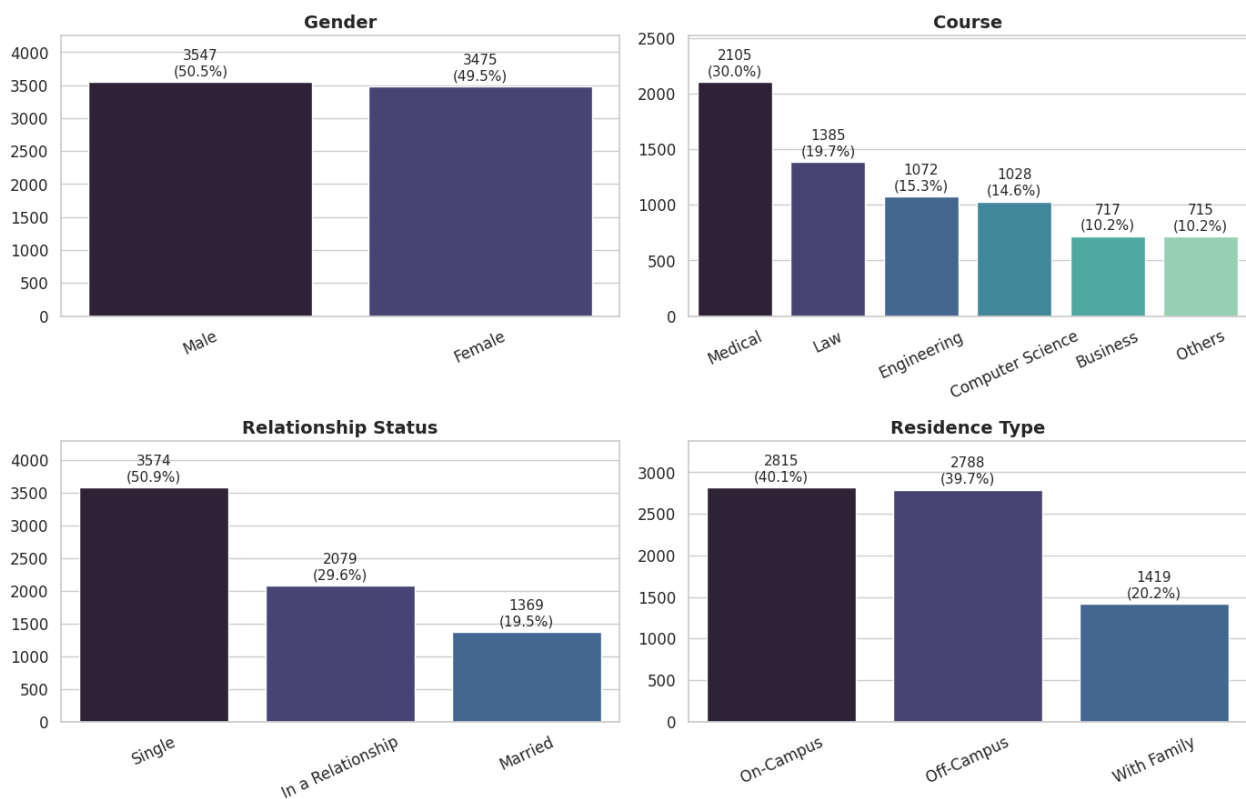


Figure 3.6. Countplot of Demographic and Background Variables

Next, let's look at the distribution of categorical variables.

The dataset is exceptionally well-balanced in terms of gender, with Males (50.5%) and Females (49.5%) having almost equal representation. This near-perfect split is ideal for analysis as it ensures that any conclusions drawn about depression risk are not biased toward a specific gender.

Students from the Medical field constitute the largest portion of the sample at 30.0%, followed by Law (19.7%). These two disciplines are often associated with high academic pressure and long study hours, which could be significant factors in mental health studies. Other majors like Engineering, Computer Science, and Business are more evenly distributed, providing a diverse academic background for the model to learn from.

A majority of the students identify as Single (50.9%), while about 29.6% are In a Relationship and 19.5% are Married. Relationship status often plays a role in emotional support systems. The presence of nearly 20% married students - likely correlating with the older "outliers" seen in the Age variable - adds an interesting dimension to how domestic responsibilities might interact with academic stress.

Around 80% of students live away from their families, split almost equally between On-Campus (40.1%) and Off-Campus (39.7%). Only 20.2% live With Family. Students living away from home (in dorms or rented rooms) may face higher risks of loneliness or financial independence stress, which are known contributors to depression among the youth.

3.7.2.6. Distribution of Lifestyle and Behavioral Variables

DISTRIBUTION OF LIFESTYLE AND BEHAVIORAL VARIABLES

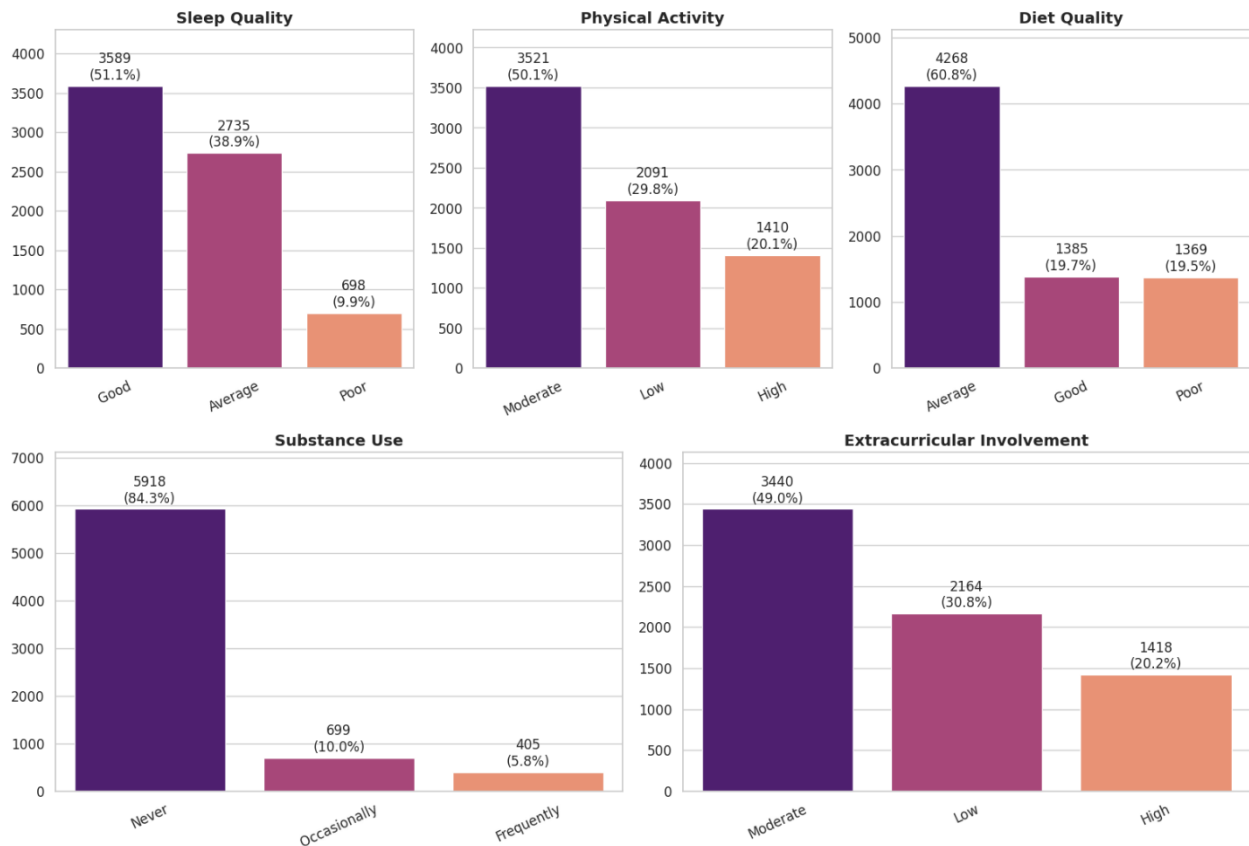


Figure 3.7. Countplot of Lifestyle and Behavioral Variables

With a group of lifestyle and behavioral variables:

Over half of the students report having Good sleep quality (51.1%), but a significant 48.9% fall into the Average or Poor categories. Sleep is a foundational pillar of mental health; students in the "Poor" category (nearly 10%) are likely the most vulnerable group. This variable often acts as a strong predictor because sleep deprivation is both a cause and a symptom of depression.

The majority of students engage in Moderate physical activity (50.1%), while 29.8% have Low activity levels. Regular exercise is a known protective factor against stress. The 30% of students with low activity might lack the "endorphin boost" that helps regulate mood, making them a key segment to monitor for depression risk.

Having 60.8% of students maintain an Average diet, while about 20% have a Poor diet. Student life is often characterized by irregular eating habits and fast food. A poor diet can lead to nutritional deficiencies that affect brain function and emotional stability, potentially exacerbating existing depressive tendencies.

While a vast majority Never use substances (84.3%), there is a small but critical 5.8% who use them Frequently. Frequent substance use is often a maladaptive coping mechanism for underlying mental health issues. This small group represents a "high-risk red flag" that the predictive model should prioritize.

Nearly half of the students have Moderate involvement (49.0%). Participation in extracurriculars generally provides social support and a sense of belonging. The 30.8% with Low involvement might be experiencing social isolation, which is a significant contributor to student depression.

3.7.2.7. Distribution of Health and Support System Variables

DISTRIBUTION OF HEALTH AND SUPPORT SYSTEM VARIABLES

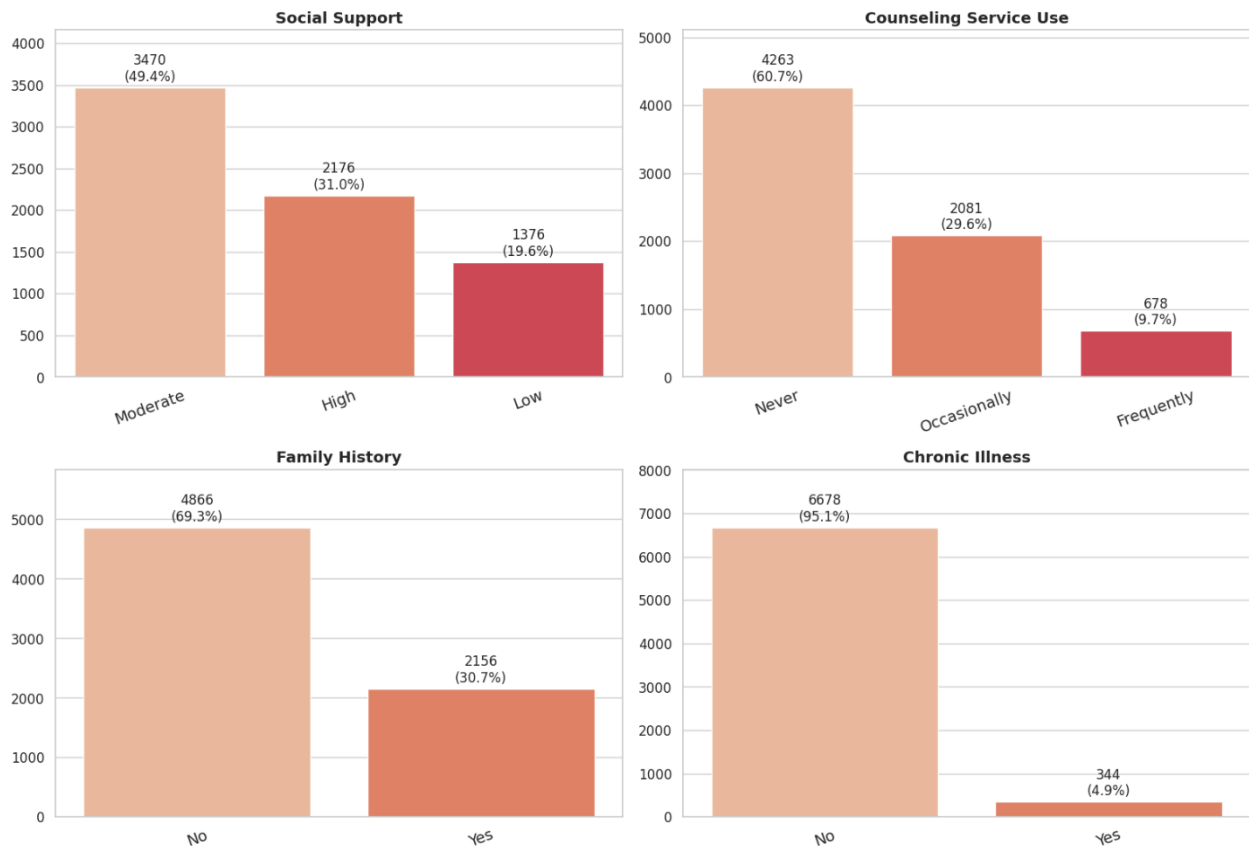


Figure 3.8. Countplot of Health and Support System variables

Nearly half of the students report having Moderate social support (49.4%), while 19.6% have Low support. Social support acts as a "buffer" against life's stressors. Students in the "Low" category are at a significantly higher risk because they lack a reliable safety net of friends or mentors to lean on during difficult times, which is a common precursor to clinical depression.

A large majority of students (60.7%) have Never used counseling services, while only 9.7% use them Frequently. This could indicate two things: either many students do not feel the need for professional help, or there is a significant barrier (stigma, cost, or lack of awareness) preventing them from seeking support. Frequent use often correlates with "High Risk" as it identifies students who are already actively struggling with their mental health.

About 30.7% of students have a Family History of mental health issues. Genetic predisposition is a well-documented risk factor for depression. Students with a "Yes" in this category may have a biological vulnerability that, when combined with academic or financial stress, significantly increases their probability of being in the "High Risk" group.

The vast majority of students do not have a Chronic Illness (95.1%), with only a small segment of 4.9% reporting "Yes". Although this group is small, it is highly critical. Dealing with a long-term physical illness while managing academic demands creates an immense double burden, which often leads to decreased emotional resilience and a higher risk of developing depressive symptoms.

3.7.2.8. Distribution of Demographic and Background Variables by Target

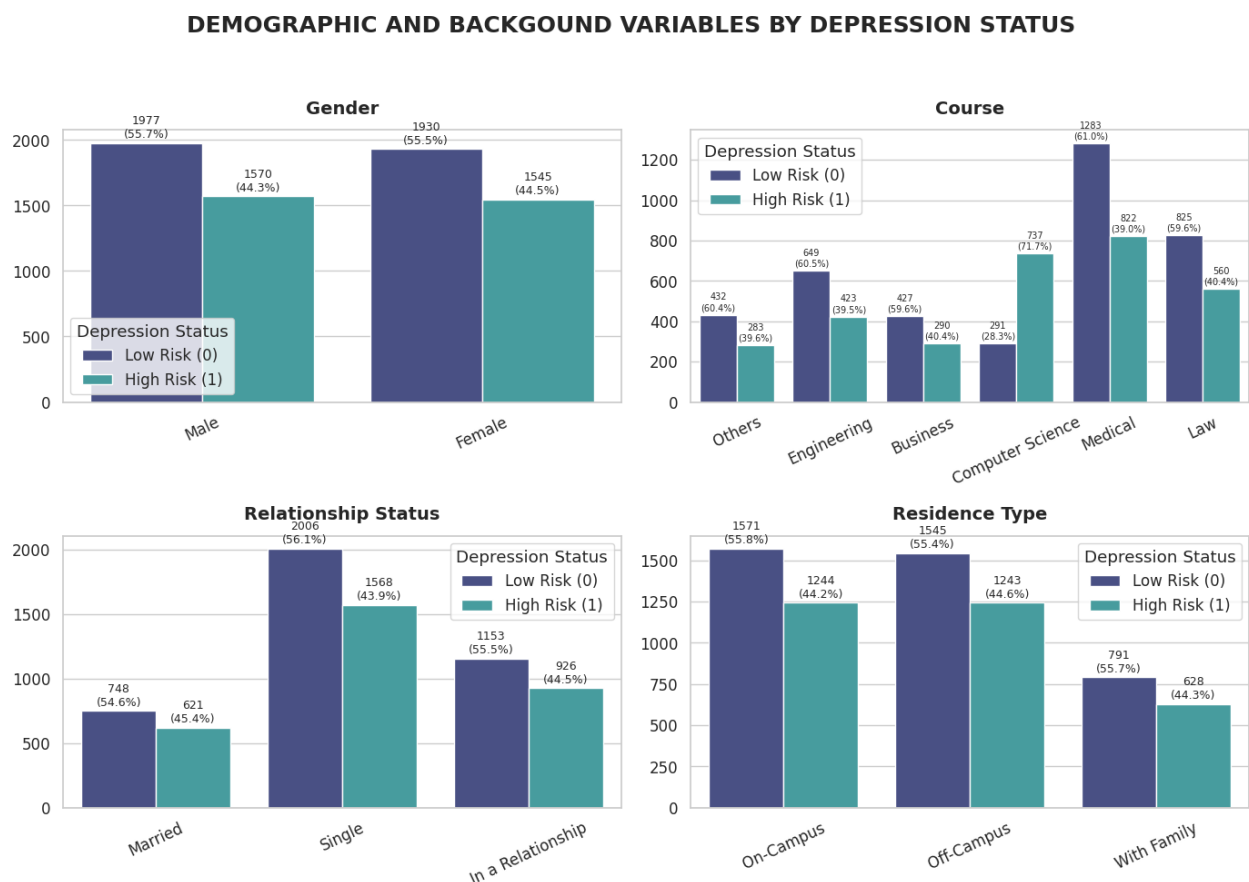


Figure 3.9. Demographic and Background Variables by Depression Status

There is no significant disparity in depression risk between Males (44.3%) and Females (44.5%). Both genders show an almost identical probability of being in the "High Risk" category. This suggests that gender is not a strong discriminating factor for depression in this dataset, indicating that the mental health challenges are experienced universally across the student body.

The Course variable reveals a striking insight: students in Computer Science have a disproportionately higher rate of High Risk (71.7%) compared to all other majors. In contrast, fields like Medical (39.0%) and Engineering (39.5%) show much lower risk rates. This high concentration in Computer Science suggests unique stressors within this field - perhaps related to screen time, isolation, or specific academic pressures - making this variable a crucial predictor for your machine learning model.

Relationship Status and Residence Type show remarkably consistent risk levels, hovering around 44-45% across all sub-categories. Whether a student is single or married, or whether they live on-campus or with family, the likelihood of depression remains stable. While these factors are often thought to influence mental health, they do not appear to be primary drivers in this specific population, as the "High Risk" group is distributed evenly regardless of living or social situation.

3.7.2.9. Distribution of Lifestyle and Behavioral Variables by Target

LIFESTYLE AND BEHAVIORAL VARIABLES BY DEPRESSION STATUS

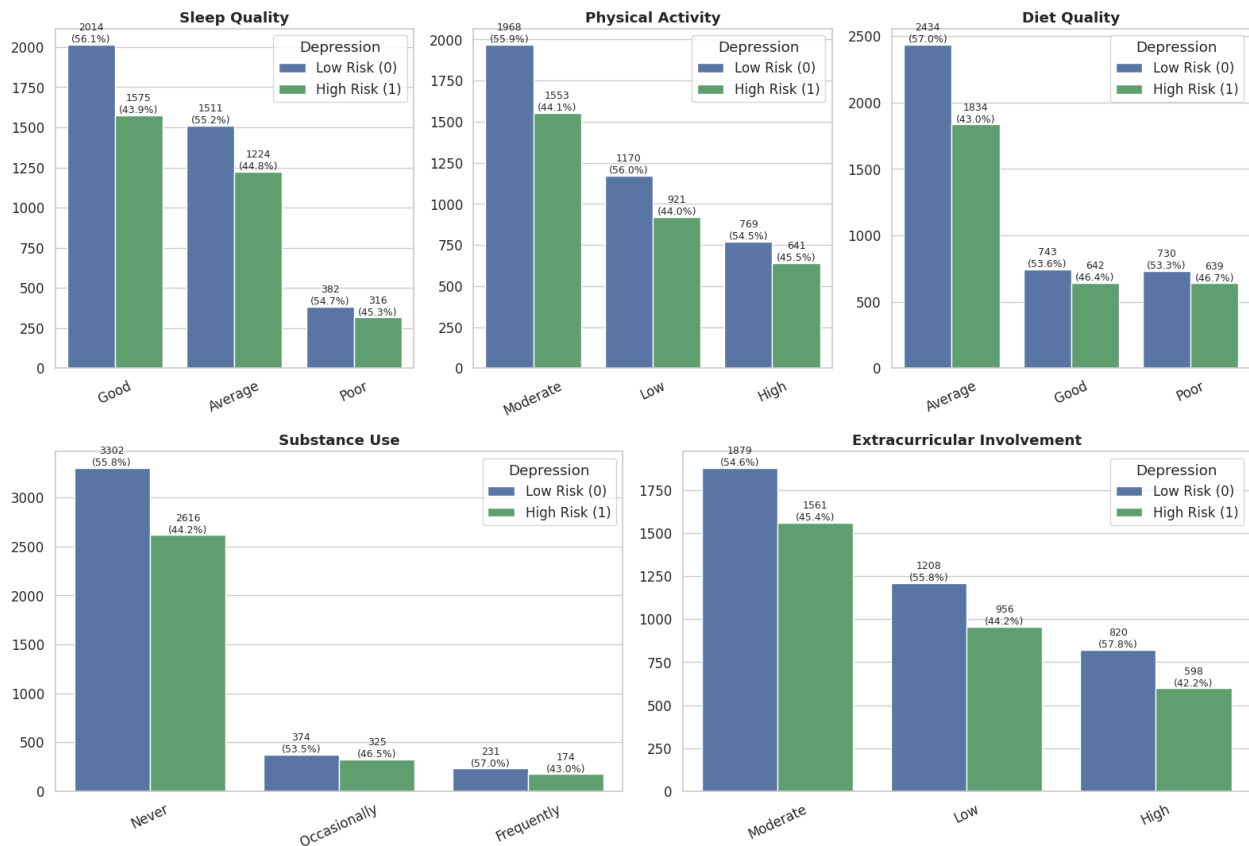


Figure 3.10. Lifestyle and Behavioral Variables by Depression Status

Variables like Sleep Quality and Physical Activity show an almost uniform distribution of depression risk. Whether a student reports "Good" or "Poor" sleep, the probability of being in the High Risk category remains remarkably stable at around 44-45%. While science usually links poor sleep to higher depression, this dataset suggests that for these students, the risk is persistent regardless of their physical habits. This implies that improving lifestyle alone might not be enough to mitigate the risk in this specific population.

Similarly, Diet Quality and Substance Use do not show the expected "spike" in depression risk for the poorer lifestyle choices. Students with a "Poor" diet or those who "Frequently" use substances exhibit a depression rate of 46.7% and 43.0% respectively - virtually indistinguishable from their healthier counterparts. This indicates that these

behavioral features have low feature importance or low predictive power on their own, as they fail to create a clear separation between the two target classes.

Even Extracurricular Involvement, often seen as a social buffer, shows a flat trend with risk levels hovering around 42-45%. This reinforces a recurring theme in your analysis: the "High Risk" group is remarkably well-integrated across almost all lifestyle categories. It suggests that student depression in this context is a complex, multi-factorial issue that isn't easily explained by simple habits.

3.7.2.10. Distribution of Health and Support System Variables by Target

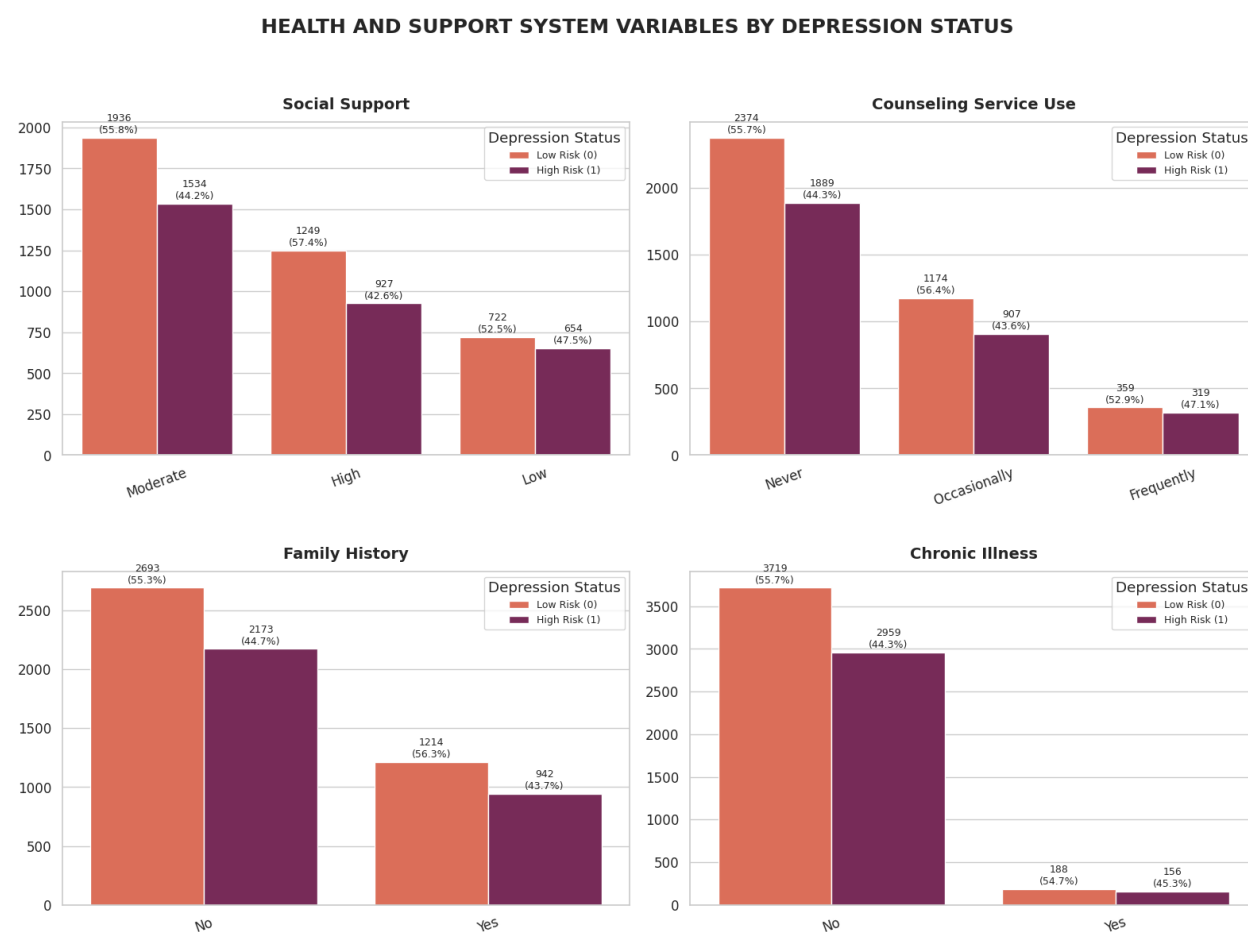


Figure 3.11. Health and Support System Variables by Depression Status

The data indicate a subtle but logical trend in the support variables. Students with Low Social Support exhibit a slightly higher depression risk (47.5%) compared to those with High support (42.6%). Students who use Counseling Services "Frequently" show a

higher risk rate (47.1%) than those who have never used them. These variables are self-reflective: those who seek frequent help or feel unsupported are naturally more likely to be struggling with their mental health. But the difference is still relatively small, suggesting that even students with high support or those who never seek counseling are not immune to high-risk status.

Having a Family History of mental health issues or a Chronic Illness does not appear to significantly increase the probability of being at high risk in this specific dataset. The depression rates remain remarkably stable between 43.7% and 45.3% regardless of these medical backgrounds. This is a counterintuitive finding, as medical literature often strongly links these factors to higher risk. It suggests that the "environmental" stress of being a student in this population may be such a dominant factor that it overshadows these biological and historical vulnerabilities.

3.7.2.11. Correlation of Numerical Variables with Target

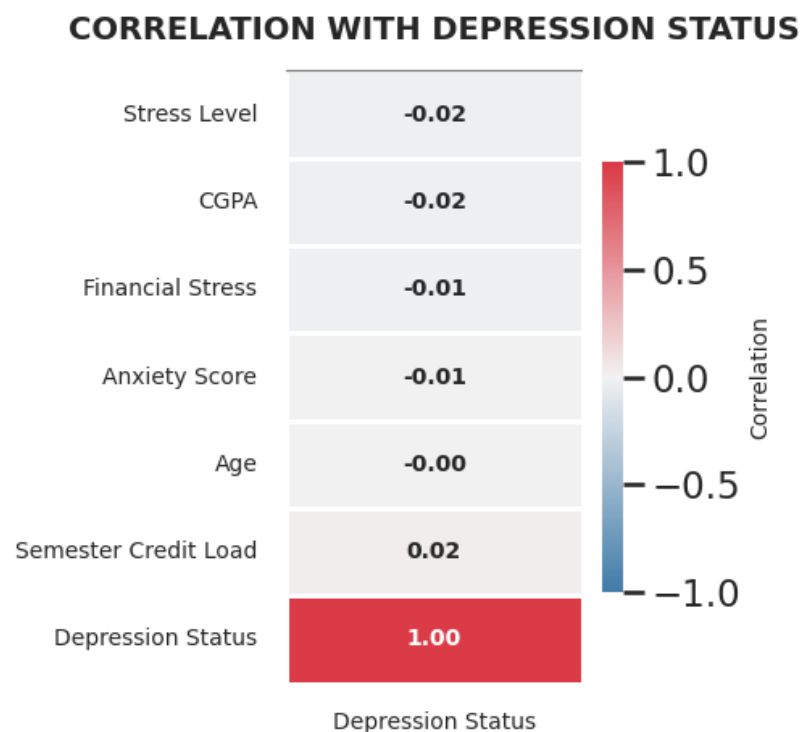


Figure 3.12. Correlation with Depression Status

The correlation heatmap illustrates the linear relationship between the numerical features and the target variable, Depression Status. All calculated coefficients for variables such as Stress Level, CGPA, Financial Stress, Anxiety Score, Age, and Semester Credit Load are remarkably close to zero, ranging from -0.02 to 0.02. This indicates a near-total absence of linear correlation within this dataset. It confirms that knowing a student's age, grades, or even reported stress levels does not provide a direct linear signal for predicting their depression risk. This is a critical finding: it suggests that the relationships in this data are either highly non-linear or that the categorical variables (like “Course”) carry far more weight in determining the final classification than these numerical metrics.

3.7.3. Transformation

The raw dataset underwent several essential transformation steps to ensure consistency and compatibility with the selected machine learning algorithms.

Text normalization was performed on all categorical variables by converting strings to lowercase and removing extraneous whitespaces, which mitigated the risk of creating redundant categories due to typographical errors.

```
# Text Normalization
for col in df.select_dtypes(include='object'):
    df[col] = df[col].str.strip().str.lower()
```

Encoding techniques were applied; binary features such as Family_History were mapped to 1 and 0, while ordinal variables like Sleep_Quality and Diet_Quality were assigned numerical scales to preserve their inherent hierarchy.

```
# Encoding Categorical Data
# Binary (Yes/No)
binary_cols = ['Family_History',
               'Chronic_Illness']
for col in binary_cols:
    df[col] = df[col].map({'yes': 1, 'no': 0})
# Ordinal
ordinal_cols = ['Sleep_Quality',
```



```

        'Physical_Activity',
        'Diet_Quality',
        'Social_Support',
        'Substance_Use',
        'Counseling_Service_Use',
        'Extracurricular_Involvement']
ordinal_mappings = {
    'Sleep_Quality': {'poor': 1, 'average': 2, 'good': 3},
    'Physical_Activity': {'low': 1, 'moderate': 2, 'high': 3},
    'Diet_Quality': {'poor': 1, 'average': 2, 'good': 3},
    'Social_Support': {'low': 1, 'moderate': 2, 'high': 3},
    'Extracurricular_Involvement': {'low': 1, 'moderate': 2, 'high': 3},
    'Substance_Use': {'never': 1, 'occasionally': 2, 'frequently': 3},
    'Counseling_Service_Use': {'never': 1, 'occasionally': 2, 'frequently':
3}
}
for col, mapping in ordinal_mappings.items():
    df[col] = df[col].map(mapping)

```

StandardScaler was utilized to normalize the continuous and encoded features, ensuring a mean of zero and a unit variance. This step was crucial for distance-based models like SVM and Logistic Regression to prevent features with larger numerical ranges from disproportionately influencing the model's coefficients.

```

from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
# Standardization applied to selected features
X_train_sel_scaled = scaler.fit_transform(X_train_selected)
X_test_sel_scaled = scaler.transform(X_test_selected)
# Standardization applied to the full feature set
X_train_full_scaled = scaler.fit_transform(X_train_full)
X_test_full_scaled = scaler.transform(X_test_full)

```

3.7.4. Splitting

The preprocessed dataset was partitioned into a Training Set (80%) and a Testing Set (20%) using a stratified sampling approach. This split was intentionally performed before

any feature selection or scaling procedures to ensure the highest level of objectivity. By isolating the test set at this early stage, the model development process remains untainted by information from unseen data, thereby preventing "data leakage" and ensuring that the final performance metrics are a true reflection of the model's generalizability.

```
from sklearn.model_selection import train_test_split
# 1. Split the original dataset
train_df, test_df = train_test_split(
    df,
    test_size=0.2,
    random_state=42,
    stratify=df['Depression_Status']
)
# 2. Separate features (X) and target (y) from each subset
# Training set
y_train = train_df['Depression_Status']
X_train_raw = train_df.drop(columns=['Depression_Status'])
# Testing set
y_test = test_df['Depression_Status']
X_test_raw = test_df.drop(columns=['Depression_Status'])
```

3.7.5. Feature Selection

After the data split, a hybrid feature selection strategy was executed exclusively on the training set to identify the most significant predictors of depression.

3.7.5.1. Chi-Square Test

A Chi-square test was applied to categorical variables to evaluate their statistical relationship with the target label, retaining those with a $p - value < 0.05$.

```
from scipy.stats import chi2_contingency
# 1. List of categorical features to be tested
cat_cols = [
    'Gender', 'Course', 'Sleep_Quality', 'Physical_Activity',
    'Diet_Quality', 'Social_Support', 'Relationship_Status',
    'Substance_Use', 'Counseling_Service_Use', 'Family_History',
```

```

    'Chronic_Illness', 'Extracurricular_Involvement',
    'Residence_Type'
]
# 2. Initialize results list
results = []
# 3. Perform Chi-square test using TRAINING DATA ONLY
for col in cat_cols:
    # Create contingency table between feature and target (train set)
    table = pd.crosstab(train_df[col], train_df['Depression_Status'])
    # Apply Chi-square test
    chi2, p, dof, expected = chi2_contingency(table)
    results.append({
        'Feature': col,
        'Chi2': chi2,
        'p-value': p
    })
# 4. Convert results to DataFrame and sort by p-value
chi_df = pd.DataFrame(results).sort_values(by='p-value')
# 5. Display results (used for feature selection)
print("CHI-SQUARE RESULTS (TRAIN SET ONLY)")
print(chi_df)

```

Table 3.3. Chi-Square Results (Train Set Only)

Feature	Chi2	p-value
Course	286.145750	9.516668e-60
Social_Support	9.928500	6.983187e-03
Diet_Quality	9.511333	8.602809e-03
Relationship_Status	3.388621	1.837259e-01
Extracurricular_Involvement	2.399465	3.012748e-01
Counseling_Service_use	1.915542	3.837473e-01
Substance_Use	1.784317	4.097704e-01
Family_History	0.490954	4.835020e-01
Physical_Activity	1.252101	5.346994e-01

Chronic_Illness	0.286398	5.925383e-01
Sleep_Quality	0.833927	6.590451e-01
Gender	0.144096	7.042426e-01
Residence_Type	0.085712	9.580491e-01

Chi-square test of independence was conducted on the training set to evaluate the statistical relationship between categorical features and Depression_Status. The analysis revealed a significant divergence in predictive power among the variables. Course emerged as the most dominant factor with an extremely low p-value ($p = 9.51 \times 10^{-60}$), indicating a powerful association between a student's academic field and their mental health status. Social Support ($p = 0.0069$) and Diet Quality ($p = 0.0086$) also demonstrated strong statistical significance at the $\alpha = 0.001$ level. These three features were prioritized for the "Selected" dataset. Other factors such as Gender ($p = 0.70$), Sleep Quality ($p = 0.65$), and Physical Activity ($p = 0.53$) failed to meet the significance threshold, suggesting that within this specific population, environmental and social factors outweigh basic demographic or physical indicators in predicting depression.

3.7.5.2. Random Forest Feature Importance

Random Forest (RF) Classifier was used to rank features by Gini Importance (Mean Decrease in Impurity). The model was trained with 200 estimators on the training set.

```
# 1. Create a copy of X_train_raw to avoid modifying the original data
X_train_tmp = X_train_raw.copy()
# 2. Encode categorical variables into numeric format (for Random Forest)
X_train_encoded = pd.get_dummies(X_train_tmp, drop_first=True)
from sklearn.ensemble import RandomForestClassifier
# Train Random Forest model on the encoded training data
rf = RandomForestClassifier(
    n_estimators=200,
    random_state=42
)
rf.fit(X_train_encoded, y_train)
```

```
# Create feature importance dataframe
feature_importance = pd.DataFrame({
    'Feature': X_train_encoded.columns,
    'Importance': rf.feature_importances_
}).sort_values(by='Importance', ascending=False)
print(feature_importance.head(10))
```

Table 3.4. Feature Importance from Random Forest Classifier

Feature	Importance
CGPA	0.137252
Age	0.096175
Semester_Credit_Load	0.095641
Anxiety_Score	0.066696
Financial_Stress	0.066316
Stress_Level	0.065775
Course_computer science	0.043222
Physical_Activity	0.039496
Extracurricular_Involvement	0.039438
Social_Support	0.036322

To provide a more intuitive visualization of the numerical results, the figure below presents a bar plot illustrating the feature importance values.

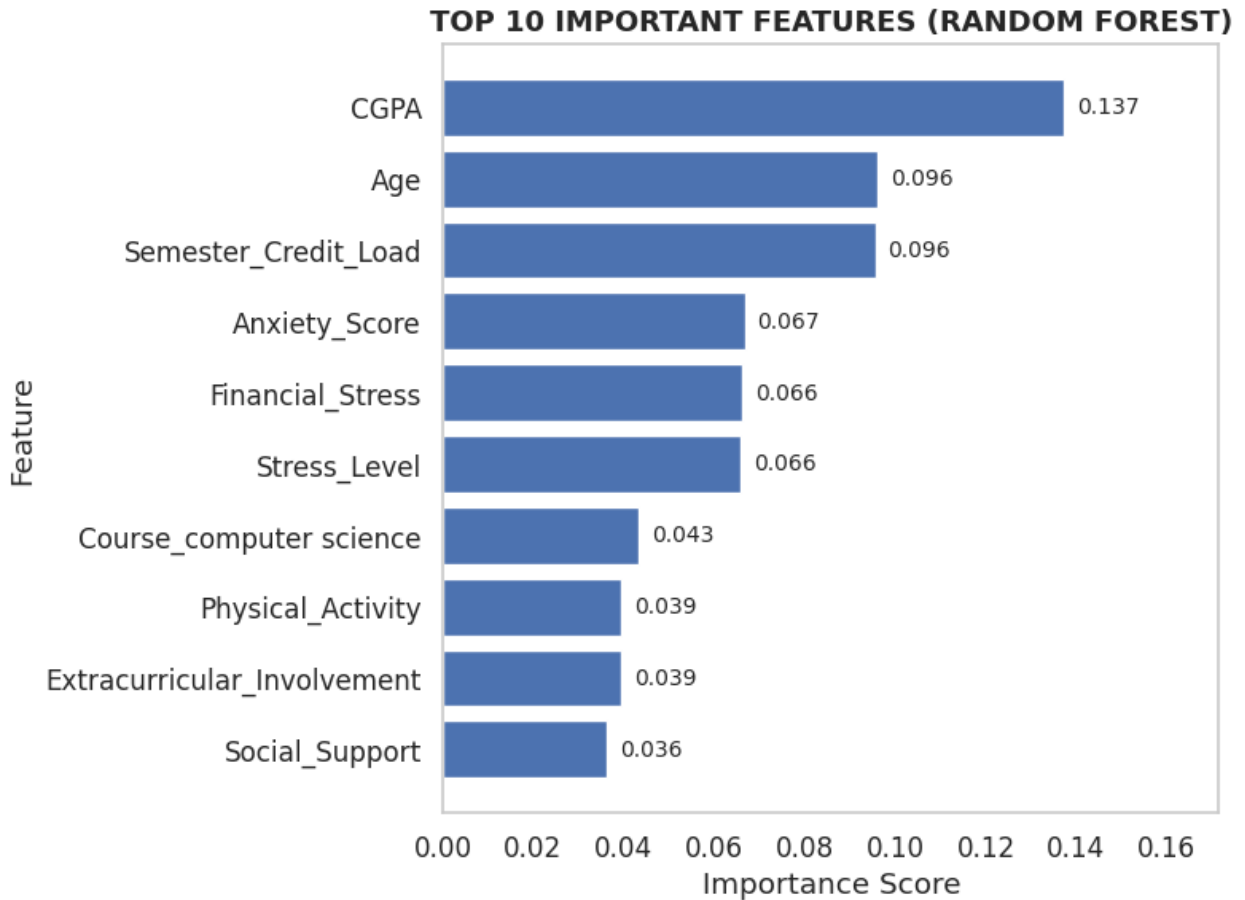


Figure 3.13. Feature Importance from Random Forest Classifier

The Random Forest Feature Importance analysis provides a clear quantitative hierarchy of the predictors. According to the results, CGPA stands out as the most dominant feature with an importance score of 0.137, indicating that academic performance is a primary driver in the model's classification of depression. This is followed by Age and Semester Credit Load, both contributing significantly with a score of 0.096. Clinical and external pressure factors, specifically Anxiety Score (0.067), Financial Stress (0.066), and Stress Level (0.066), also demonstrate substantial predictive power. Based on these rankings, the top six features - from Stress Level upwards - were selected as they represent the most impactful numerical and clinical indicators in the dataset.

By synthesizing these six high-impact features with the three statistically significant variables previously identified through the Chi-square test (Course, Social Support, and Diet Quality), a final refined set of 9 core features was established. This "Selected" feature set represents a robust multi-dimensional approach, combining categorical statistical

significance with numerical predictive importance. This synthesis ensures that the final models are trained on the most relevant environmental, academic, and psychological drivers while minimizing noise from redundant variables.

CHAPTER 4: RESULTS AND DISCUSSION

4.1. Overview of Experimental Setup

The empirical phase of this study is formulated as a supervised binary classification problem, with the primary objective of predicting students' mental health status, specifically the presence of depressive symptoms. The target variable is categorized into two distinct classes: High Risk (Yes), representing students exhibiting significant depressive indicators, and Low Risk (No), representing those without such symptoms.

Given the sensitive nature of mental health prediction, it is essential to employ models that not only achieve strong predictive performance but also provide meaningful insights into the underlying factors. So, three machine learning algorithms were selected, each representing a different modeling philosophy and offering complementary strengths.

Logistic Regression (LR) is employed as the baseline model. Despite being a linear model, Logistic Regression is inherently designed for classification tasks through the use of the Sigmoid function, which transforms input values into probabilities within the range of 0 to 1. The model is particularly valuable in this study due to its high interpretability. Specifically, it enables the estimation of coefficients and corresponding odds ratios, allowing researchers to quantify how various factors - such as academic pressure, lifestyle, or social support - affect the likelihood of a student being classified as high-risk. This interpretability is crucial in mental health research, where understanding why a prediction is made is often as important as the prediction itself. But its primary limitation lies in the assumption of linear relationships between features and the log-odds of the outcome, which may not adequately capture the complex and non-linear nature of psychological data.

To address this limitation, the Support Vector Machine (SVM) is introduced as a more flexible classification approach. SVM operates by identifying an optimal hyperplane that maximizes the margin between the two classes in a high-dimensional feature space. One of its key strengths lies in its ability to handle high-dimensional data effectively, making it suitable for datasets with numerous behavioral and psychological indicators. Through the use of kernel functions (such as the radial basis function), SVM can transform the original feature space into a higher-dimensional space, enabling the separation of non-linearly distributed data. This makes SVM particularly useful for capturing complex patterns that linear models may fail to detect. SVM requires careful feature scaling and parameter tuning, and it can be sensitive to noise in the data.

Random Forest (RF) is employed as an ensemble learning method to enhance predictive robustness and stability. Unlike individual decision trees, which are prone to overfitting, RF constructs a large number of trees using different subsets of the data and aggregates their predictions through a majority voting mechanism. This bagging approach effectively reduces variance and improves generalization performance. Random Forest is especially well-suited for this study because it can capture non-linear relationships and intricate interactions between variables, such as the combined influence of academic stress and sleep quality on mental health. It is relatively robust to outliers and noise, and it provides measures of feature importance, which further supports interpretability from a different perspective.

The combination of these three models allows for a comprehensive evaluation framework, balancing interpretability, flexibility, and predictive performance in the context of mental health classification.

4.2. Results

In addition to the full feature dataset, a reduced dataset is constructed based on features selected from the Chi-square test and Random Forest classifier. The following code is used to generate this selected dataset, which is then encoded and aligned between the training and testing sets to ensure consistency for subsequent model training.

```
# Get original feature names (for mapping)
all_original_cols = X_train_raw.columns.tolist()
# Select top important features from Random Forest
rf_top = feature_importance.head(6) ['Feature']
rf_selected = set()
# Map encoded (dummy) features back to original features
for feat in rf_top:
    for col in all_original_cols:
        if feat.startswith(col):
            rf_selected.add(col)
print("Selected features from RF:", rf_selected)
# Combine with Chi-square selected features
chi_selected = ['Course', 'Diet_Quality', 'Social_Support']
final_features = list(set(chi_selected) | rf_selected)
print("\nFinal selected features (SELECTED):", final_features)
```

```

# --- Create SELECTED dataset ---
# Subset original data with selected features
X_train_selected_raw = X_train_raw[final_features]
X_test_selected_raw = X_test_raw[final_features]
# Encode categorical variables
X_train_selected = pd.get_dummies(X_train_selected_raw, drop_first=True)
X_test_selected = pd.get_dummies(X_test_selected_raw, drop_first=True)
# Align test set with training set
X_test_selected =
X_test_selected.reindex(columns=X_train_selected.columns, fill_value=0)
print(f"\nSELECTED dataset shape: Train {X_train_selected.shape}, Test
{X_test_selected.shape}")

```

Final selected features include: Age, Course, Anxiety_Score, Social_Support, CGPA, Stress_Level, Financial_Stress, Diet_Quality, and Semester_Credit_Load.

The shape of the train set is (5617, 13), the test set (1405, 13).

4.2.1. Performance on the Selected Dataset

The following tables present the performance results focusing on Class 1 (High Risk), as this class represents students with depressive symptoms and is the primary concern of this study.

Table 4.1. Performance on the Selected dataset

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.6285	0.71	0.27	0.39
SVM	0.6263	0.70	0.28	0.40
Random Forest	0.6014	0.57	0.39	0.46

Overall, Logistic Regression and SVM achieve similar performance, with accuracy values of 0.6285 and 0.6263, respectively. But Random Forest shows slightly lower accuracy at 0.6014.

In terms of precision for the high-risk class, Logistic Regression (0.71) and SVM (0.70) outperform Random Forest (0.57), indicating that their positive predictions are more

reliable. However, all models exhibit relatively low recall values for Class 1. Logistic Regression and SVM achieve recall values of 0.27 and 0.28, respectively, while Random Forest performs slightly better at 0.39.

This suggests that although Random Forest is less precise, it is more effective at identifying high-risk students. On the other hand, Logistic Regression and SVM tend to miss a large proportion of actual high-risk cases, as reflected in their low recall.

The results indicate a trade-off between precision and recall across the models. While Logistic Regression and SVM provide more accurate positive predictions, Random Forest demonstrates a better ability to detect high-risk individuals, which is particularly important in the context of mental health prediction.

4.2.2. Performance on the Full Dataset

Table 4.2. Performance on the Full Dataset

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.6278	0.71	0.27	0.39
SVM	0.6206	0.67	0.28	0.40
Random Forest	0.6142	0.62	0.34	0.44

Table 4.2 presents the performance of the three models using the full feature set. Logistic Regression achieves the highest accuracy (0.6278), followed closely by SVM (0.6206), while Random Forest attains a slightly lower accuracy of 0.6142. The performance differences among the models are relatively small, indicating that all three approaches perform at a comparable level when using the full dataset.

In terms of precision for the high-risk class, Logistic Regression maintains the highest value (0.71), followed by SVM (0.67) and Random Forest (0.62). This suggests that Logistic Regression provides the most reliable positive predictions when identifying students at risk.

Similar to the results observed in the selected dataset, recall for Class 1 remains relatively low across all models. Logistic Regression and SVM achieve recall values of 0.27 and 0.28, respectively, while Random Forest again shows a slightly higher recall at 0.34. This indicates that Random Forest is more effective in capturing high-risk cases, although at the expense of lower precision.

The results on the full dataset reveal a consistent trade-off between precision and recall. While Logistic Regression offers more precise predictions, Random Forest demonstrates a relatively better ability to identify at-risk students. This pattern is consistent with the findings from the selected feature set, suggesting that the choice of model has a greater impact on recall performance than the size of the feature set.

4.2.3. Comparison of Feature Sets and Models

The results indicate that the performance differences between the selected and full feature sets are relatively minor. Accuracy values remain nearly unchanged across both settings for all models, suggesting that the reduced feature subset is sufficient to capture most of the predictive information. This implies that the feature selection process effectively removes redundant or less informative variables without significantly degrading model performance.

In terms of recall for the high-risk class, the improvement from using the full dataset is limited. Random Forest shows a slight decrease in recall when moving from the selected dataset (0.39) to the full dataset (0.34), while Logistic Regression and SVM maintain nearly identical recall values. This further supports the observation that feature selection has minimal impact on the models' ability to detect high-risk individuals.

When comparing the models, Logistic Regression and SVM consistently achieve higher precision, indicating that their positive predictions are more reliable. Both models suffer from low recall, meaning that a significant number of high-risk cases are not identified. But Random Forest demonstrates relatively higher recall across both datasets, suggesting that it is more effective at capturing high-risk students, albeit with lower precision.

Generally, the results highlight a clear trade-off between precision and recall among the models. Logistic Regression and SVM prioritize prediction accuracy, while Random Forest provides a better balance toward identifying high-risk cases. This trade-off is particularly important in the context of mental health prediction, where detecting at-risk individuals is often more critical than minimizing false positives.

These findings suggest that model selection plays a more critical role than feature selection in determining recall performance.

4.2.4. Feature Importance Analysis using Logistic Regression

To further understand the influence of individual factors on mental health outcomes, LR is employed for feature importance analysis. Due to its interpretability, this model allows for direct examination of the relationship between each feature and the probability of being classified as high-risk. The features are ranked based on the absolute values of their coefficients to identify the most influential variables. Table 4.3 presents the detailed results of this analysis.

```
# Ensure Logistic Regression is trained on SELECTED dataset
lr_model = models["Logistic Regression"]
lr_model.fit(X_train_sel_scaled, y_train)
# Create feature importance table (coefficient & odds ratio)
importance_df = pd.DataFrame({
    'Feature': X_train_selected.columns,
    'Coefficient': lr_model.coef_[0],
    'Odds_Ratio': np.exp(lr_model.coef_[0]) # e^beta
})
# Compute absolute coefficient for ranking
importance_df['Abs_Coefficient'] = importance_df['Coefficient'].abs()
importance_df = importance_df.sort_values(by='Abs_Coefficient',
ascending=False)
# Display results
print("--- DETAILED FEATURE ANALYSIS (LOGISTIC REGRESSION) ---")
print(importance_df[['Feature', 'Coefficient',
'Odds_Ratio']].reset_index(drop=True))
```

Table 4.3. Detailed Feature Analysis (Logistic Regression)

Feature	Coefficient	Odds Ratio
Course_computer science	0.432816	1.541593
Course_medical	-0.080281	0.922857
Social_Support	-0.057422	0.944196
Semester_Credit_Load	0.056865	1.058513
Course_others	-0.051935	0.949390
Anxiety_Score	0.040169	1.040986
CGPA	-0.038553	0.962181
Course_engineering	-0.033761	0.966803
Financial_Stress	-0.030798	0.969672
Course_law	-0.023668	0.976610
Diet_Quality	0.016758	1.016899
Stress_Level	0.015559	1.015680
Age	-0.008297	0.991737

The results reveal that the most significant predictor in this dataset is the field of study, particularly for students majoring in Computer Science. With a positive coefficient of 0.4328 and an Odds Ratio of 1.5416, the model indicates that Computer Science students are approximately 1.54 times, or 54%, more likely to be classified as high-risk compared to the baseline group, assuming all other factors remain constant. This finding strongly validates the theoretical framework proposed by (Lipson et al., 2022), which identifies high-intensity STEM disciplines as primary environments for academic burnout and psychological distress.

Beyond the specific academic major, several other variables contribute significantly to the probability of depression risk. The Semester Credit Load, with an Odds Ratio of 1.0585, suggests that each additional credit hour registered increases the likelihood of being

high-risk by approximately 5.8%. This quantifiable link between credit overload and mental health underscores the tangible impact of academic pressure. Psychological indicators such as Anxiety Score (OR = 1.0410) and Stress Level (OR = 1.0157) also show a positive correlation with depression risk, confirming that pre-existing mental strain is a reliable predictor for more severe depressive states. These factors, when combined, create a profile of academic and psychological vulnerability that the model effectively captures.

Several features act as protective factors, as indicated by coefficients below zero and Odds Ratios less than one. Social Support stands out as a crucial mitigator of risk; with an Odds Ratio of 0.9442, higher levels of perceived support are associated with a 5.6% decrease in the odds of being high-risk. This highlights the vital role of interpersonal relationships and community in buffering students against mental health challenges. Academic performance, represented by CGPA (OR = 0.9622), also serves as a slight protective factor, suggesting that academic success may either provide a sense of achievement that mitigates symptoms or that students with lower depressive risks are naturally better positioned to maintain higher grades. Other factors, such as Diet Quality and specific majors like Medical or Law also show protective tendencies in this dataset, providing a comprehensive view of the diverse elements that influence student well-being.

4.2.5. Classification Performance and Error Analysis

To gain a deeper understanding of the model's diagnostic capability beyond simple accuracy, a Confusion Matrix is utilized. This evaluation tool is essential for identifying specific classification errors, categorized into False Positives (Type I errors) and False Negatives (Type II errors). In the context of mental health screening, distinguishing between these errors is critical; for instance, a False Negative - where a high-risk student is incorrectly classified as low-risk - could result in a lack of timely psychological support. The following heatmap (Figure X) visualizes the performance of the LR model on the selected feature set, comparing the actual labels against the model's predictions.

```
from sklearn.metrics import confusion_matrix
# Generate predictions
```

```

y_pred_lr = lr_model.predict(X_test_sel_scaled)
# Compute confusion matrix
cm = confusion_matrix(y_test, y_pred_lr)
# Plot heatmap
plt.figure(figsize=(6, 4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', cbar=False)
# Title and labels
plt.title('Confusion Matrix - Logistic Regression (Selected
Features)'.upper(),
          fontsize=13, fontweight='bold')
plt.xlabel('Predicted', fontsize=12)
plt.ylabel('Actual', fontsize=12)
# Show plot
plt.show()

```

CONFUSION MATRIX - LOGISTIC REGRESSION (SELECTED FEATURES)

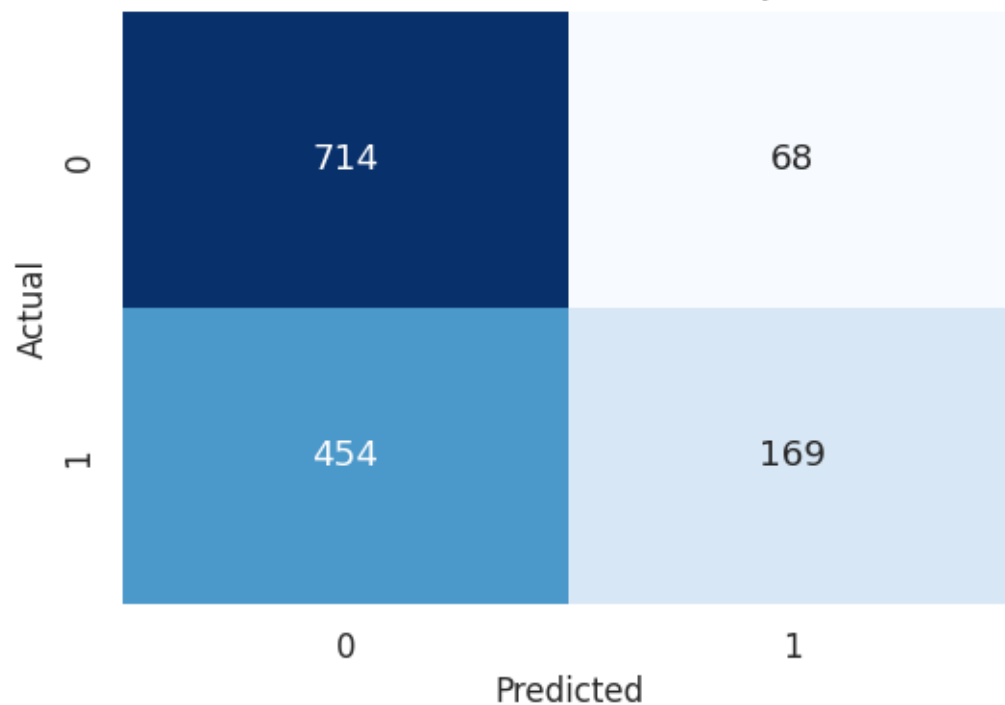


Figure 4.1. Confusion Matrix – Logistic Regression

The Confusion Matrix provides a detailed breakdown of the Logistic Regression model's predictions on the testing set, revealing both its strengths and its critical limitations in clinical screening. Out of 1,405 total observations, the model correctly identified 714

True Negatives (TN) and 169 True Positives (TP). While the overall accuracy remains moderate, a closer look at the error distribution reveals a significant discrepancy between the model's ability to identify low-risk versus high-risk individuals.

The model demonstrates high performance in identifying students at low risk for depression. With only 68 False Positives (FP), the model maintains high Specificity, meaning it rarely misclassifies healthy students as "High-risk." This conservative approach ensures that university counseling resources are not overwhelmed by "false alarms." However, this comes at a high cost: the False Negatives (FN) stand at 454 cases. This indicates that a large proportion of students who actually possess high-risk indicators were incorrectly flagged as low-risk by the model.

In the context of mental health screening, a high FN rate is a significant concern. It suggests that while the Logistic Regression model is "safe" (avoiding over-diagnosis), it is not sufficiently "sensitive" to capture the subtle or complex patterns of depression in nearly 72.8% of high-risk cases. This performance gap suggests that while demographic and academic features (like being a Computer Science student) are strong indicators, they may not be exhaustive enough for a purely linear model to capture the full spectrum of psychological distress.

4.2.6. ROC Curve

To provide a comprehensive evaluation of the model's discriminative ability, the Receiver Operating Characteristic (ROC) curve is analyzed. While the previous Confusion Matrix provided a snapshot of performance at a fixed threshold, the ROC curve illustrates the diagnostic power of the Logistic Regression model across all possible classification thresholds. This is achieved by plotting the True Positive Rate (Sensitivity) against the False Positive Rate (1-Specificity).

```
from sklearn.metrics import roc_curve, roc_auc_score
# Compute predicted probabilities (15)
y_prob_lr = lr_model.predict_proba(X_test_sel_scaled)[:, 1]
# Calculate ROC metrics
```

```

fpr, tpr, _ = roc_curve(y_test, y_prob_lr)
auc_score = roc_auc_score(y_test, y_prob_lr)
# Plot ROC curve
plt.figure(figsize=(7, 5))
plt.plot(fpr, tpr, color='darkorange', lw=2, label=f'ROC curve (AUC =
{auc_score:.2f})')
plt.plot([0, 1], [0, 1], color='navy', lw=2, linestyle='--', label='Random
baseline')
# Axis and labels
plt.xlim([0.0, 1.0])
plt.ylim([0.0, 1.05])
plt.xlabel('False Positive Rate', fontsize=12)
plt.ylabel('True Positive Rate', fontsize=12)
plt.title('ROC Curve'.upper(),
          fontsize=13, fontweight='bold')
# Legend and grid
plt.legend(loc="lower right")
plt.grid(alpha=0.3)
# Show plot
plt.show()

```

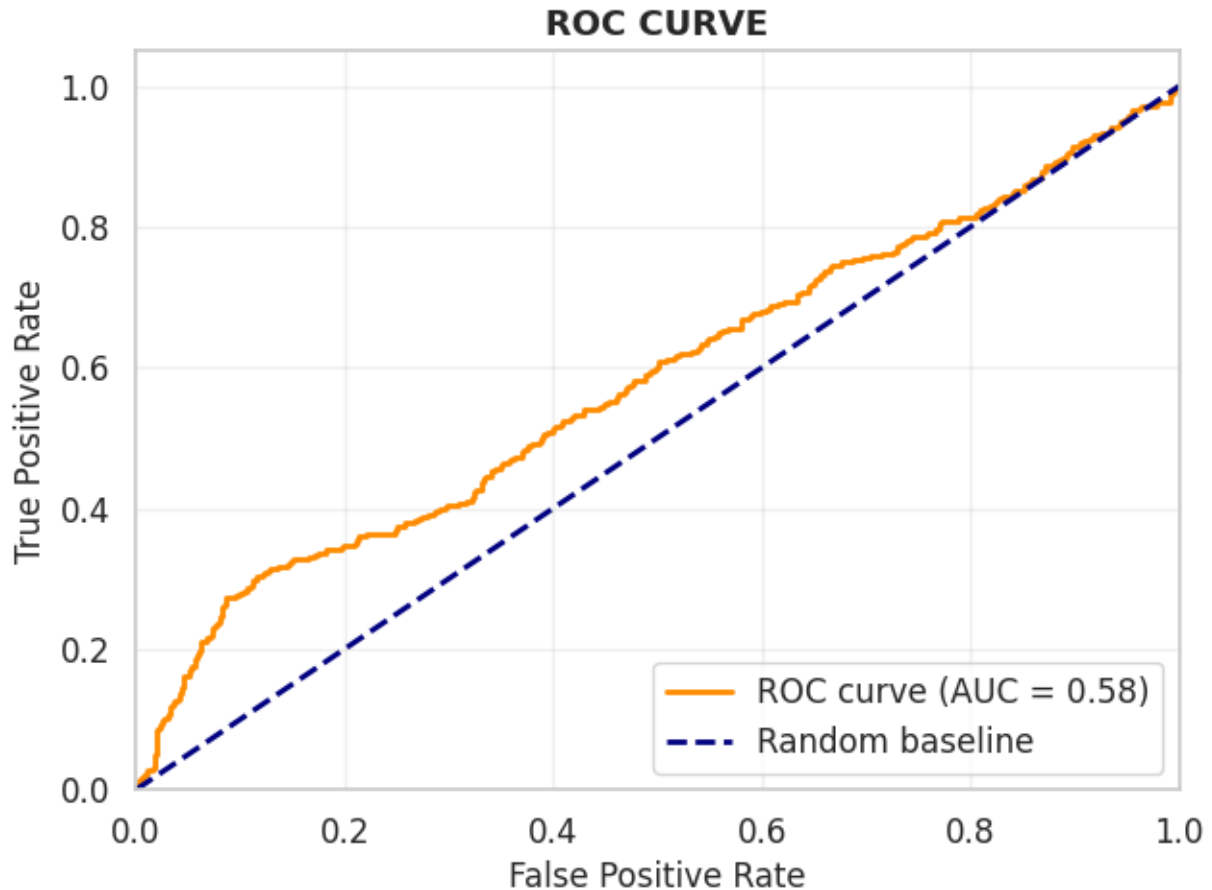


Figure 4.2. Operating Characteristic (ROC) Curve

The ROC curve provides a visual representation of the Logistic Regression model's ability to discriminate between high-risk and low-risk students across all possible classification thresholds. The plot displays the trade-off between the True Positive Rate (Sensitivity) and the False Positive Rate (1-Specificity). In this specific experiment, the model yielded an AUC (Area Under the Curve) of 0.58. Mathematically, an AUC of 0.50 represents a model with no predictive power, equivalent to random guessing (represented by the diagonal navy dashed line). Therefore, a score of 0.58 indicates that the model is only marginally better than a coin flip at distinguishing between the two psychological states.

The shape of the orange curve, which stays significantly close to the random baseline, suggests that the model struggles to maintain a high True Positive Rate without

simultaneously increasing the False Positive Rate. This lack of "curvature" toward the top-left corner of the plot is a clear indicator of low discriminative power. In the context of student mental health, this implies that the linear combinations of variables like "Course Type" or "Stress Level" do not provide a sufficiently clear boundary to separate individuals effectively using a standard Logistic Regression framework. While the model is statistically significant in identifying certain risk factors, its overall diagnostic performance on the test set is considered poor.

4.2.7. The Robustness of the Logistic Regression

To evaluate the reliability of the Logistic Regression model, a comparison between Training Accuracy and Testing Accuracy was conducted. This analysis is essential to detect Overfitting, a common issue where a model performs well on training data but fails to generalize to unseen observations.

```
# Compute accuracy on train and test sets
train_acc = lr_model.score(X_train_sel_scaled, y_train)
test_acc = lr_model.score(X_test_sel_scaled, y_test)
# Print results
print(f"Train accuracy: {train_acc:.4f}")
print(f"Test accuracy: {test_acc:.4f}")
```

The results yielded a Training Accuracy of 61.79% and a Testing Accuracy of 62.85%. The fact that the testing accuracy is slightly higher than the training accuracy, with a negligible gap of approximately 1%, indicates that the model possesses high stability and excellent generalization capabilities. There is no evidence of overfitting; instead, the model successfully captured the underlying patterns of the dataset without "memorizing" specific noise. However, the consistent accuracy around the 62% mark suggests a state of Underfitting, likely due to the linear nature of Logistic Regression. This implies that while the model is stable, it may require more complex, non-linear algorithms to fully resolve the intricate psychological correlations within student mental health data.

4.3. Discussion

The empirical results of this study provide a nuanced understanding of the factors contributing to student depression and the technical challenges in predicting mental health outcomes. The discussion focuses on three key dimensions: the significance of academic

discipline, the inherent trade-offs in model performance, and the complexity of psychological data.

The feature importance analysis via Logistic Regression highlights Computer Science as the most potent predictor of high-risk status (OR = 1.5416). This finding suggests that students in tech-heavy, high-intensity STEM fields face unique psychological pressures, possibly stemming from heavy workloads, high competition, and social isolation. The protective role of Social Support (OR = 0.9442) underscores that interpersonal networks act as a vital buffer against academic stress. Interestingly, the slight protective effect of CGPA (OR = 0.9622) implies that academic success may bolster a student's self-esteem and resilience, or alternatively, that lower psychological distress allows for better academic focus.

A critical observation in this study is the consistent trade-off between Precision and Recall. Logistic Regression and SVM proved to be highly conservative; while their "High Risk" predictions were reliable (Precision ~ 0.71), they missed nearly 73% of actual cases (Recall ~ 0.27). But Random Forest demonstrated a superior ability to "catch" more at-risk individuals (Recall up to 0.39), albeit with more false alarms. In the context of university mental health screening, this suggests that ensemble models like Random Forest might be more appropriate if the priority is early intervention, where "missing" a student in need is considered more costly than a false positive.

The modest overall accuracy ($\sim 62\%$) and AUC (0.58) reflect the profound complexity of human psychology. Unlike physical health data, depression is influenced by a multitude of latent variables - such as personal history, genetic predisposition, and current life events - which were not fully captured in the survey. The fact that the model showed no overfitting (Train accuracy $\approx$ Test accuracy) confirms that the results are stable and generalizable. The state of Underfitting suggests that linear frameworks are insufficient to resolve the intricate, non-linear relationships of mental distress. This paves the way for future research to utilize more advanced deep learning architectures or longitudinal data to improve predictive sensitivity.

CHAPTER 5: CONCLUSION

5.1. Summary

The primary objective of this research was to identify the key determinants of student depression and evaluate the efficacy of machine learning models in predicting high-risk individuals. Through the analysis of various academic and psychological factors, several significant conclusions can be drawn:

Academic discipline is a profound predictor of mental health status. Students majoring in Computer Science exhibit a significantly higher risk of depression compared to other fields, with an Odds Ratio of 1.54. This suggests that the high-intensity environment of tech-heavy STEM subjects contributes to increased psychological strain. Additionally, Semester Credit Load and Anxiety Score were identified as primary risk-increasing factors, while Social Support and Academic Success (CGPA) acted as vital protective buffers.

Second, regarding model performance, the study reveals a clear trade-off between interpretability and sensitivity. While Logistic Regression (LR) and SVM provided highly reliable positive predictions (High Precision), they suffered from low Recall, failing to identify a large proportion of at-risk students. Random Forest (RF), despite lower precision, demonstrated superior robustness in capturing high-risk cases (Higher Recall). The overall accuracy of approximately 60-63% and the lack of overfitting confirm that while the models are stable and generalizable, the complexity of student mental health likely requires more advanced, non-linear approaches for higher predictive accuracy.

5.2. Recommendations

5.2.1. For the University and Faculty Administration

The University should prioritize mental health resources for high-risk departments, specifically Computer Science and STEM-related faculties. Special counseling sessions or stress-management workshops tailored to the specific pressures of these majors are essential.

Academic advisors should closely monitor students with exceptionally high semester credit loads, as this was identified as a quantifiable stressor. Implementing "Mental Health Days" or flexible deadlines during peak midterm seasons could mitigate burnout.

5.2.2. For the Student Counseling Center

The Counseling Center should consider integrating machine learning models, specifically Random Forest, as a preliminary screening tool. Due to its higher Recall, it can act as a "first-pass" filter to identify students who may need urgent attention, which can then be followed up by professional clinical interviews.

Since Social Support is a significant protective factor, the university should foster peer-mentoring programs and student clubs to reduce social isolation, especially for freshmen and students in high-pressure majors.

5.3. Limitations and Future Research

Despite the significant insights, this study possesses certain limitations. The dataset relies on self-reported surveys, which may introduce subjectivity or social desirability bias. The 62% accuracy indicates that there are latent variables - such as genetic factors or specific personal life events - not captured in the current model.

Future research should focus on incorporating longitudinal data to track mental health changes over time rather than a single snapshot. Exploring Deep Learning architectures (such as Neural Networks) or Hybrid Models may help resolve the non-linear complexities that standard algorithms like Logistic Regression struggled to capture.

REFERENCES

- American Psychological Association. (2022). *Mental health of college students is a growing concern*. American Psychological Association. <https://www.apa.org/monitor/2022/10/mental-health-college-students>
- Breiman, L. (2001). Random Forests. *Machine Learning*, 45(1), 5–32. <https://doi.org/10.1023/A:1010933404324>
- Cortes, C., & Vapnik, V. (1995). Support-vector networks. *Machine Learning*, 20(3), 273–297. <https://doi.org/10.1007/BF00994018>
- Cox, D. R. (1958). The Regression Analysis of Binary Sequences. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 20(2), 215–232. <https://doi.org/10.1111/j.2517-6161.1958.tb00292.x>
- Eisenberg, D., Hunt, J., & Speer, N. (2013). Mental Health in American Colleges and Universities. *Journal of Nervous & Mental Disease*, 201(1), 60–67. <https://doi.org/10.1097/NMD.0b013e31827ab077>
- Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning*. Springer New York. <https://doi.org/10.1007/978-0-387-84858-7>
- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An Introduction to Statistical Learning*. Springer US. <https://doi.org/10.1007/978-1-0716-1418-1>
- Li, Z., Zhou, H., Xu, Z., & Ma, Q. (2025). Machine learning and public health policy evaluation: research dynamics and prospects for challenges. *Frontiers in Public Health*, 13. <https://doi.org/10.3389/fpubh.2025.1502599>
- Lipson, S. K., Zhou, S., Abelson, S., Heinze, J., Jirsa, M., Morigney, J., Patterson, A., Singh, M., & Eisenberg, D. (2022). Trends in college student mental health and help-seeking by race/ethnicity: Findings from the national healthy minds study, 2013–2021. *Journal of Affective Disorders*, 306, 138–147. <https://doi.org/10.1016/j.jad.2022.03.038>

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., VanderPlas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, E. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.

Rajkomar, A., Dean, J., & Kohane, I. (2019). Machine Learning in Medicine. *New England Journal of Medicine*, 380(14), 1347–1358. <https://doi.org/10.1056/NEJMra1814259>

Shatte, A. B. R., Hutchinson, D. M., & Teague, S. J. (2019). Machine learning in mental health: a scoping review of methods and applications. *Psychological Medicine*, 49(09), 1426–1448. <https://doi.org/10.1017/S0033291719000151>

Topol, E. J. (2019). High-performance medicine: the convergence of human and artificial intelligence. *Nature Medicine*, 25(1), 44–56. <https://doi.org/10.1038/s41591-018-0300-7>

World Health Organization. (2022). World mental health report: Transforming mental health for all. In *Journal of the Royal Statistical Society Series B: Statistical Methodology* (Vol. 20, Number 2). Oxford University Press (OUP). <https://doi.org/10.1111/j.2517-6161.1958.tb00292.x>

World Health Organization. (2023). *Depressive disorder (depression)*. Oxford University Press (OUP). World Health Organization. <https://doi.org/10.1111/j.2517-6161.1958.tb00292.x>

Yin, S., Li, Z., Song, D., Mu, H., Niu, Y., & Wang, X. (2024). Study on the Energy Evolution Law and Bursting Liability of Coal Failure with Different Joint Inclination Angles. *Applied Sciences*, 14(3), 1120. <https://doi.org/10.3390/app14031120>